

REVIEW

Role of Potassium Ion Channels in Vascular Tone Regulation and Hypertension

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ABSTRACT: Potassium ion channels are critical regulators of vascular homeostasis and cardiac electrophysiology that act by regulating the resting membrane potential via hyperpolarization or depolarization. They are modulated in response to intracellular or extracellular voltage or ion concentrations, which is key in preserving vascular tone and cardiac contractility. In vascular smooth muscle cells, large-conductance calcium-activated potassium channels are influenced by both intracellular calcium ion concentrations and during depolarization, whereas small- and intermediate-conductance channels are strictly dependent on intracellular calcium ions only, which in turn are regulated by ryanodine receptors of the sarcoplasmic reticulum. On the other hand, voltage-dependent potassium channels are activated by membrane depolarization, and the resting membrane potential is restored via negative feedback. Genome-wide association studies in humans identified variation in genes encoding voltage-dependent and 2-pore domain potassium ion channels as being associated with the risk of developing hypertension. In rodent models of hypertension, arterial expression and activity of large-conductance calcium-activated potassium channels are upregulated, whereas voltage-dependent, ATP-sensitive, and inward-rectifier potassium channels are downregulated. In cardiomyocytes, potassium currents play a key role in regulating cardiac action potential and refractory periods, with their dysregulation contributing to arrhythmogenesis. Given the therapeutic significance of potassium ion channels in antiarrhythmic treatments, we have reviewed their potential to exhibit blood pressure-lowering effects using evidence from human and animal studies. More research is warranted to investigate the significance of existing drugs, including amiodarone and sotalol, in the treatment of hypertension.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: calcium ■ endothelial cells ■ hypertension ■ membrane potentials ■ muscle, smooth, vascular

Potassium ion (K^+) channels facilitate the movement of K^+ ions across cellular membranes in response to the electrochemical gradient,^{1,2} playing a key role in several physiological processes.³ These channels form when K^+ channel transmembrane proteins assemble into highly selective pores and are critical for cardiovascular function.¹

Potassium channels, including calcium-activated channels, voltage-gated (K_v), ATP-sensitive potassium (K_{ATP}), inward-rectifier potassium (K_{ir}), and 2-pore domain potassium (K_{2P}) channels, within vascular cells regulate vascular homeostasis and tone^{4–8} and within cardiomyocytes regulate cardiac contractility and rhythm (Figure 1).^{8,9} These channels modulate membrane potential (E_m), calcium ion (Ca^{2+}) influx to regulate vascular smooth muscle cell (VSMC) contractility in arteries.⁷ E_m is the electrical potential difference across the cell membrane generated

by unequal ion distribution and selective permeability.¹⁰ In the context of hypertension, activation of K^+ channels promotes its efflux to induce membrane hyperpolarization, a shift towards more negative E_m , which consequently suppresses VGCC (voltage-gated Ca^{2+} channel) activity to reduce further Ca^{2+} influx to maintain vascular tone and promote vasodilation.^{1,3,7,11} Vascular tone is the degree of contractile activity of VSMCs within the walls of small arteries and arterioles. It plays a key role in regulating blood pressure by determining the vessel diameter and thereby peripheral resistance.² In atrial cardiomyocytes, K^+ channels are essential for the generation and propagation of the action potential (AP), a rapid and transient change in E_m produced by coordinated ion fluxes, which enables synchronized myocardial contraction.^{8,12–15} Within the AP, depolarization refers to the rapid rise in E_m caused primarily by inward-moving cation currents,

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
AP	action potential
APD	action potential duration
BK/BK_{Ca}	large-conductance calcium-activated potassium channel
Ca²⁺	calcium ions
CYP11B2	cytochrome P450 family 11 subfamily B member 2
EC	endothelial cells
EDH	endothelium-dependent hyperpolarization
E_m	membrane potential
eNOS	endothelial NO synthase
IK/K_{Ca}3.1/KCNN4	intermediate-conductance calcium-activated channels
I_{Kr}	rapid delayed rectifier potassium current
I_{Ks}	slow delayed rectifier potassium current
I_{Kur}	ultrarapid delayed rectifier potassium current
K⁺	potassium ions
K_{2P}	2-pore domain potassium channels
K_{ATP}	ATP-sensitive potassium channel
KCNJ11	gene encoding ATP-sensitive channel subunit 6.2
KCNJ8	gene encoding ATP-sensitive channel subunit 6.1
K_{ir}	inward-rectifier potassium channel
K_v	voltage-gated potassium channels
K_v1.5 (KCNA5)	voltage-gated potassium channel encoded by the KCNA5 gene
Na⁺	sodium ions
PKA	Protein kinase A
SHR	spontaneously hypertensive rats
SK	small-conductance calcium-activated potassium channels
SK2/K_{Ca}2.2/KCNN2	pore-forming α-subunit of the small-conductance calcium-activated potassium channels
SK3/K_{Ca}2.3/KCNN3	small-conductance calcium-activated potassium channel subunit 2.3
SNP	single-nucleotide polymorphism
SUR2	sulfonylurea receptor subunit 2
TRPV4	transient receptor potential vanilloid 4 channels
VE-cadherin	vascular endothelial cadherin
VGCC	voltage-gated Ca ²⁺ channels
VSMC	vascular smooth muscle cells
WKY	Wistar-Kyoto rats

Highlights

- Potassium ion channels are central regulators of vascular tone and cardiac electrophysiology, primarily by modulating membrane potential through hyperpolarization and depolarization mechanisms that control calcium ion influx and cellular excitability.
- Potassium channel subtypes such as large-conductance calcium-activated, voltage-gated potassium channel, ATP-sensitive potassium channel, inward-rectifier potassium channel, and 2-pore domain potassium channels in vascular smooth muscle cells and endothelial cells maintain vascular homeostasis through highly regulated calcium ion-dependent signaling pathways.
- Dysregulation of potassium channels contributes to hypertension by promoting membrane depolarization, increased intracellular calcium ion, sustained vasoconstriction, endothelial dysfunction, and pathological vascular remodeling.
- Evidence from animal and human studies indicates upregulation of large-conductance calcium-activated channels (likely compensatory) and downregulation or impaired function of voltage-gated potassium channels, ATP-sensitive potassium channels, and inward-rectifier potassium channels, leading to altered vascular tone.
- Genetic studies (genome-wide association studies and Mendelian randomization) and pharmacological evidence highlight potassium channels as promising therapeutic targets, although current potassium channel modulating drugs, such as antiarrhythmics, show limited and nonspecific antihypertensive effects, warranting further targeted research.

whereas repolarization is the restoration of the negative E_m largely mediated by outward K⁺ currents.⁸ The cardiac AP duration (APD), the period from depolarization to repolarization, is essential for maintaining normal heart rhythm. APD determines the refractory period, the phase after depolarization when cardiomyocytes are unresponsive to electrical stimulus, thereby ensuring orderly impulse propagation and rhythmic timing of myocardial contractions.^{8,16,17} Due to these essential roles, impaired function or regulation of K⁺ channel activity has been linked to several cardiovascular pathologies such as cardiac arrhythmias, hypertension, atherosclerosis, stroke, and coronary artery disease, as well as metabolic disorders that contribute to vascular dysfunction.^{9,18,19}

Current studies are focused on identifying new therapeutic significance of K⁺ channels in cardiovascular diseases other than arrhythmias. However, the broad expression of K⁺ channels in most cell types within the body has been a major therapeutic challenge due to off-target effects. A comprehensive review of ion channels in the cardiovascular system is available elsewhere.²⁰ This review provides an overview of the mechanisms through

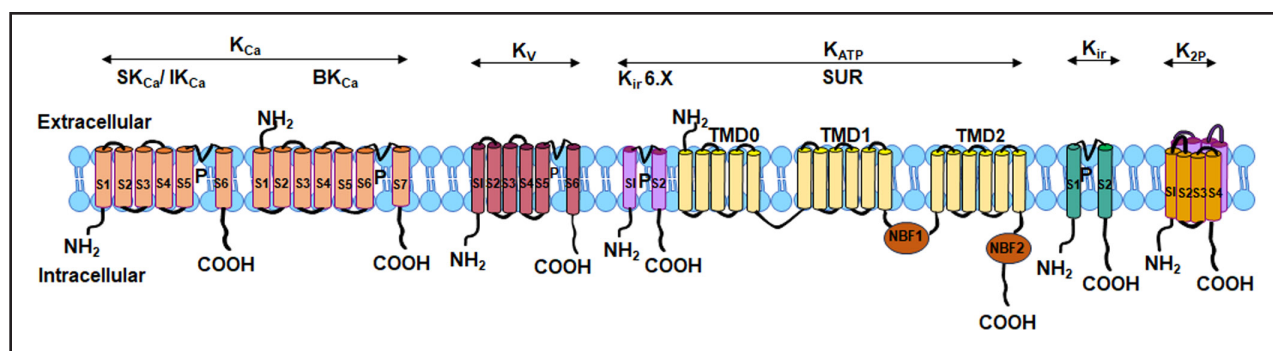


Figure 1. Schematic representation illustrating the structure of vascular potassium ion channels.

Calcium ion-activated potassium channel (K_{Ca}) channels: Small/intermediate-conductance calcium-activated potassium channels (SK_{Ca}/IK_{Ca}) consisting of 6-transmembrane (TM) spanning domains (S1–S6) in the α -subunits that form the channel, pore-forming P loop, and cytosolic N and C termini. Large-conductance calcium-activated potassium channel (BK_{Ca}) has 7-TM domains, the pore-forming domain between (S5–S6) and a C-terminal cytosolic region. Voltage-gated potassium channel (K_v) channels: voltage-dependent potassium channels contain 6 TM (S1–S6) segments, a pore-forming P loop, and intracellular N and C termini. ATP-sensitive potassium channel (K_{ATP}) channels: K_{ATP} channels are a hetero-octameric complex composed of a 6.X subunit of ATP-sensitive potassium channel ($K_{ir}6.X$) subunit and regulatory SUR (sulfonyleurea receptor) subunits. The inward-rectifier potassium channel (K_{ir}) subunit consists of 2 TM helices S1 and S2 pore-forming region containing the pore-helix and intracellular N and C termini. The SUR subunit comprises 3 TM domains (TMD0, TMD1, TMD2) and 2 nucleotide-binding domains (NBF), extracellular N and intracellular C termini. K_{ir} channels: K_{ir} channels contain 2 TM regions (S1 and S2), a pore-forming loop (P), and cytosolic N and C termini. Two-pore domain potassium channels (K_{2P}) channels: K_{2P} α -subunits possess 4 TM domains, and 2-pore-forming loops arranged in tandem. The functional channel is formed by the dimerization of 2 α -subunits. COOH indicates N/carboxy-terminus.

which K^+ channels regulate vascular tone and their role in hypertension. We also discussed data from animal and human studies, genome-wide association studies, and Mendelian randomization studies that highlight the role of K^+ channels in hypertension. Finally, we discussed the potential of K^+ channel modulating antiarrhythmic drugs as potential therapeutic agents for the management of hypertension. Relevant publications were identified by searching the PubMed database using the following key words: “Potassium channels,” “ K^+ ion channels,” “ BK_{Ca} ,” “ IK_{Ca} ,” “ SK_{Ca} ,” “ K_{ATP} ,” “ K_v ,” “ K_{ir} ,” “ K_{2P} ,” “vascular smooth muscle cells,” “VSMC,” “endothelial,” “macrophage,” “fibroblast,” “cardiomyocytes,” “arrhythmia,” “blood pressure,” and “hypertension.” Both animal and human studies were considered for inclusion, and no language or time restrictions were applied. Studies were selected based on reporting K^+ channel expression or function, including protein expression, gene expression, or electrophysiological activity (whole-cell currents, membrane capacitance, channel conductance, or tissue-level responses assessed using organ bath experiments). Studies not reporting K^+ channel expression or functional activity were excluded.

PHYSIOLOGICAL ROLE OF K^+ CHANNELS

Endothelial Cells

Endothelial cells (ECs) form an epithelial monolayer, the endothelium, that lines the luminal surface of blood vessels.²¹ The endothelium is integral to vascular homeostasis and acts as a selective barrier to regulate the exchange of signaling molecules and cells from blood to the underlying interstitium and cells,^{21,22} thereby

playing a key role in regulating the coagulation response to vascular injury,²³ vascular tone,²⁴ and permeability of endothelial tight junctions and caveolae in response to mechanical and chemical stimuli.^{25–27} Endothelial vascular tone regulation is primarily mediated by NO,^{24,28} prostacyclin,^{24,29} and endothelium-dependent hyperpolarization (EDH) regulation.³⁰ Changes in shear stress or receptor mediated processes lead to an increase in EC intracellular Ca^{2+} levels.^{31,32} The elevated intracellular Ca^{2+} then binds to calmodulin and activates eNOS (endothelial NO synthase), which catalyzes the synthesis of NO.³³ The synthesized NO diffuses to VSMC to activate soluble guanylyl cyclase to produce cyclic guanosine monophosphate.^{34–36} This cascade of events ultimately reduces intracellular Ca^{2+} concentrations and inhibits myosin light-chain kinase activity, promoting VSMC relaxation and vasodilation.^{21,37} Furthermore, NO inhibits platelet aggregation and adhesion on the vascular wall,^{38,39} as well as leukocyte adhesion and vascular inflammation.⁴⁰ EDH is initiated when agonists such as acetylcholine activate G-protein-coupled receptors located in EC membrane, which leads to an increase in intracellular Ca^{2+} via inositol 1,4,5-trisphosphate-mediated release of Ca^{2+} from the endoplasmic reticulum and Ca^{2+} influx through TRPV4 (transient receptor potential vanilloid 4) channels (Figure 2A).⁴¹ This rise in intracellular Ca^{2+} activates the small-conductance calcium-activated channel (SK_{Ca} – $SK3$) and intermediate-conductance calcium-activated channel ($IK_{Ca}/K_{Ca}3.1/KCNN4$), resulting in K^+ efflux, endothelial hyperpolarization and EDH-mediated vasodilation in resistant arteries and arterioles.^{42,43} This hyperpolarizing current spreads to the adjacent VSMC through myoendothelial junctions and results in the

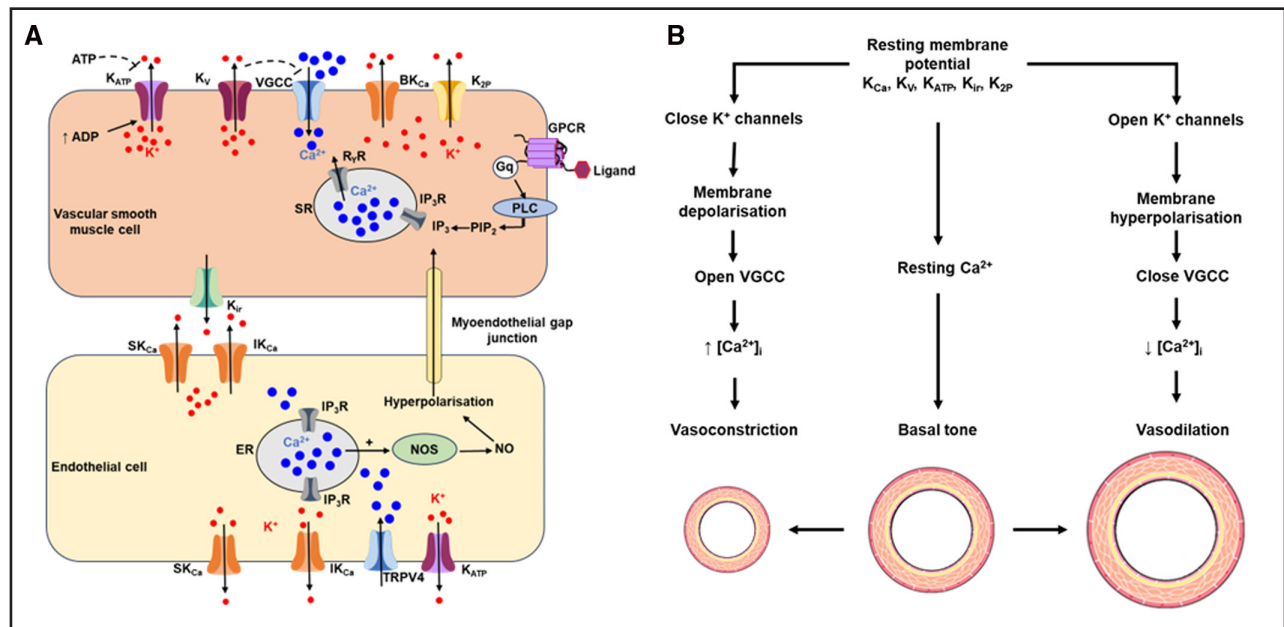


Figure 2. Role of potassium channels in modulation of vascular cell membrane potential and vascular tone.

A, In vascular smooth muscle cell (VSMC), activation of ATP-sensitive potassium channels (K_{ATP}), voltage-gated potassium channels (K_V), and 2-pore domain potassium channels (K_{2P}) leads to potassium ion (K^+) efflux, resulting in membrane hyperpolarization. This decrease in membrane potential inhibits VGCCs (voltage-gated calcium ion channels) and lowers cytosolic calcium ion (Ca^{2+}) levels, leading to VSMC relaxation. Furthermore, activation of large-conductance calcium-activated potassium channel (BK_{Ca}) by highly localized release of Ca^{2+} , mediated by RyRs (ryanodine receptors) in the sarcoplasmic reticulum (SR), results in membrane hyperpolarization and VGCC closure. As a result, downstream activation of GPCRs (G-protein-coupled receptors) triggers the production of diacylglycerol and inositol 1,4,5-trisphosphate (IP_3) via PLC (phospholipase C) and phosphatidylinositol 4,5-bisphosphate (PIP_2), leading to calcium mobilization from the SR. This Ca^{2+} wave activates K_{Ca} channels, promoting VSMC relaxation. In endothelial cells, activation of $K_{Ca2.3}/K_{Ca3.1}$ channels occurs in response to IP_3R (inositol 1,4,5-trisphosphate receptor)-mediated Ca^{2+} release from the ER after receptor stimulation (GPCR) or in response to Ca^{2+} influx through TRPV4 (transient receptor potential vanilloid 4) channels. This leads to hyperpolarization of the endothelial cell membrane. The hyperpolarization can be communicated to adjacent smooth muscle cells through myoendothelial gap junctions, resulting in a decrease in Ca^{2+} entry via VGCCs. This reduction in Ca^{2+} entry causes vasodilation. An increase in intracellular Ca^{2+} would elevate NO, which can also hyperpolarize the adjacent VSMC. **B**, Vascular tone is largely influenced by $[Ca^{2+}]_i$, which is primarily regulated by the membrane potential and the activity of VGCCs. Membrane depolarization activates VGCC, resulting in increased Ca^{2+} influx and increased $[Ca^{2+}]_i$, which leads to vasoconstriction. Conversely, hyperpolarization causes the closure of VGCC, leading to reduced Ca^{2+} entry and ultimately vasodilation. $[Ca^{2+}]_i$ indicates intracellular Ca^{2+} ; Gq, G-protein q; $K_{Ca2.3}/SK_{Ca1}$, small-conductance calcium-activated potassium channel; $K_{Ca3.1}/IK_{Ca1}$, intermediate-conductance calcium-activated potassium channel; K_{Ca1} , Ca^{2+} -activated potassium channel; K_{ir} , inward-rectifier potassium channel; and NOS, NO synthase.

closure of VGCC in VSMC, thus reducing Ca^{2+} influx and consequently inducing VSMC relaxation and vasodilation (Figure 2B).⁴⁴ Furthermore, K_{ATP} channels in the endothelium have been shown to contribute to the regulation of blood flow in the coronary vasculature.⁴⁵

Vascular Smooth Muscle Cells

VSMCs are present in the medial layer of the blood vessels and exhibit remarkable elasticity.⁴⁶ Their primary function is to regulate vascular tone and blood pressure by altering the vessel diameter.⁴⁷ They can control the pulsatility of the arteries through modulating the diameter and stiffness of these vessels, depending on whether they adopt a contractile or synthetic phenotype. VSMCs undergo phenotype switching along a spectrum of contractile to synthetic states in response to external cues such as vascular injury and inflammation.^{48–50} The contractile phenotype is characterized by the expression of α -smooth muscle actin,

smooth muscle myosin heavy chain, smooth muscle 22 α , calponin proteins, and contractile VSMCs are largely quiescent.⁵¹ In contrast, synthetic VSMCs are characterized by reduced contractile protein expression and increased proliferation, migration, protein synthesis, and extracellular matrix production. In healthy vessels, the predominant VSMC population is contractile, with a small proportion of synthetic VSMCs contributing to vessel homeostasis and repair.⁵² In proliferative VSMCs, expression of IK_{Ca} ($K_{Ca3.1}/KCNN4$) and voltage-gated potassium channels encoded by the *KCNA3* gene channels is increased and serves to modulate E_m , thereby facilitating Ca^{2+} entry through non-selective cation pathways (including TRP channels) that promote proliferation and gene expression changes.^{53–55} Unlike contractile VSMCs, proliferative VSMCs have downregulated VGCCs, so these ion channels are not primarily involved in electromechanical coupling or regulation of vascular tone.^{53–55} In contrast, expression of SK_{Ca} (SK3) in normal VSMCs appears more restricted, and

IK_{Ca} expression is upregulated during vascular injury or restenosis.⁵⁶

In healthy human VSMC, K^+ ion channels are critical for regulating E_m and vascular tone through Ca^{2+} -dependent contraction and relaxation.⁵⁷ The regulation of vascular tone involves a complex interaction of vasodilatory and vasoconstrictive signals from circulating hormones, neurotransmitters, endothelium-derived factors, and blood pressure.² VSMCs integrate these signals to modulate contractility, blood vessel diameter, and vascular resistance. Hyperpolarization occurs when additional K^+ channels open or when inward depolarizing currents, such as Na^+ or Ca^{2+} , decrease.³ This reduces the intracellular Ca^{2+} concentrations, leading to closure of VGCC. Lower Ca^{2+} levels diminish the activation of calmodulin and myosin light-chain kinase, resulting in reduced phosphorylation of myosin light chains.³ Consequently, the reduced interaction between myosin and actin filaments leads to VSMC relaxation and vasodilation.⁵⁸ In contrast, K^+ channel closure or extracellular K^+ accumulation results in depolarization, leading to opening of VGCC, which increases Ca^{2+} entry and promotes VSMC contraction and vasoconstriction.^{18,58}

This dynamic interplay between K^+ -mediated E_m shifts and Ca^{2+} signaling underlies the cyclical regulation of vascular tone across the circulatory system. Various K^+ channels orchestrate the regulation of E_m via different stimuli, which is essential for normal vascular function. Large-conductance calcium-activated potassium channels (BK_{Ca}) respond to membrane depolarization and elevated intracellular Ca^{2+} , thus activating repolarization.^{59,60} Healthy VSMC predominantly express BK_{Ca} channels,⁶¹ whereas proliferative VSMC have been shown to express IK_{Ca} channels.⁵³ Similarly, K_v channels respond to depolarization, accelerating repolarization to restore the resting E_m .^{18,62,63} Voltage-gated potassium channel encoded by the *KCNA5* gene ($K_v1.5$) contributes to metabolic and cAMP-mediated vasodilation in the VSMCs.^{64,65} K_{ATP} channels open during metabolic stress, such as low ATP/ADP ratio, and are also opened through PKA (protein kinase A)-dependent phosphorylation in response to vasodilators that stimulate Gs (stimulatory G protein)-coupled receptors.^{62,66} K_{ir} channels are activated by membrane hyperpolarization and elevation in extracellular K^+ . They importantly amplify and stabilize membrane hyperpolarization initiated by other vascular K^+ channels, thereby promoting vasorelaxation.⁶⁷ Some members of K_{2P} channels that are expressed in cerebral and peripheral arteries are capable of conducting outward currents that facilitate hyperpolarization.⁶⁸ However, direct evidence for a major role of K_{2P} channels in regulating systemic vascular tone under basal conditions is limited, with functional impacts demonstrated more clearly in specific vascular beds such as cerebral and pulmonary arteries.^{69–71} The details of these potassium channels are briefly summarized in Table 1.

Cardiomyocytes

Cardiomyocytes are classified into atrial, ventricular, and pacemaker cells, each with specialized roles in cardiac contraction and rhythm.¹³³ K^+ channels play a pivotal role in maintaining sinus rhythm in the atria,¹³³ making them crucial targets in the management of arrhythmias.¹⁴ Electrophysiological activity of the atria is driven by several families of K^+ channels.^{8,12,13} These channels differ across atrial regions in terms of kinetics, gating, and distribution, as well as between epicardium and endocardium.^{8,12–15}

K_v channels provide the major repolarizing influence,^{8,12–15} including voltage-gated potassium channel encoded by the *KCND3* gene ($K_v4.3$) and voltage-gated potassium channel encoded by the *KCNA4* gene ($K_v1.4$), which are responsible for early phase 1 repolarization, as well as $K_v1.5$ (*KCNA5*), which is atrial-predominant and a key determinant of atrial APD, and voltage-gated potassium channel encoded by the *KCNH2* gene ($K_v1.1$) together with voltage-gated potassium channel encoded by the *KCNQ1* gene ($K_v7.1$) and its accessory subunit KCNE1 contribute to mid-to-late repolarization.^{8,9} Inward-rectifier potassium channels subunit 2.1 (*KCNJ2*) help maintain the resting E_m and contribute to repolarization that is required for subsequent contractility. Ligand-gated channels such as acetylcholine-activated inward-rectifier potassium channels subunit 3.1/inward-rectifier potassium channels subunit 3.4 (*KCNJ3/KCNJ5*), ATP-sensitive 6.X subunit of ATP-sensitive potassium channel channels with sulfonylurea receptor subunits (*KCNJ11* or *KCNJ8* with *ABCC9*), Ca^{2+} -activated $K_{Ca}2.2/K_{Ca}2.3$ (*KCNN2/KCNN3*), and Na^+ -activated K^+ channels modulate repolarization in response to parasympathetic tone, intracellular Ca^{2+} , and Na^+ level.^{8,12–15} TWIK (tandem of P-domains in a weak inward rectifying potassium channel)-related acid-sensitive potassium channel 1, a member of the 2-pore domain potassium (*KCNK3*), provides an additional repolarizing influence during the AP plateau (phase 2).^{8,12–15} However, the distribution of these currents varies depending on the region. For example, inward-rectifier potassium channel subunit 2.1 is present in the atrial region but not in the sinoatrial node. On the other hand, the voltage-gated potassium channel encoded by the *KCND3* gene, the voltage-gated potassium channel encoded by the *KCNA4* gene, $K_{Ca}2.2$, and $K_{Ca}2.3$ are prevalent throughout the atria.^{8,12,13,15,134} The details of these potassium channels are briefly summarized in Table 2.

PATHOPHYSIOLOGICAL ROLE OF K^+ CHANNELS

EC and VSMCs

Dysregulation of K^+ channels in EC and VSMCs has been implicated in several cardiovascular diseases, including hypertension, atherosclerosis, carotid artery disease,

Table 1. Potassium Channel Families and Their Subtypes

Fam-ily	Sub-type	Syn-onyms	Gene name (human)	No. of TM domains	Expressing vascular cells*	Functions	Agonists	Antagonists	References
K _{Ca}	K _{Ca} 2.3	Small conductance calcium activated potassium channels/ SK _{Ca}	<i>KCNN3</i>	6	EC, platelets	Activation of SK _{Ca} and IK _{Ca} triggered by increased intracellular Ca ²⁺ and activated channels promotes relaxation of EC by facilitating the efflux of K ⁺ , inducing membrane hyperpolarization.	1-EBIO, NS309, SKA-31	Apamin, UCL1684	72–75
	K _{Ca} 3.1	Intermediate conductance calcium activated potassium channels/ IK _{Ca}	<i>KCNN4</i>		EC, VSMC, macrophages, fibroblasts, T-lymphocytes, neutrophils	SK _{Ca} and IK _{Ca} contribute to endothelium-dependent vasodilation. BK _{Ca} channels coordinate with ryanodine receptors to maintain arterial tone via regulating intracellular Ca ²⁺ levels and sensing membrane potential changes. Relatively small number of BK _{Ca} channels can have a significant effect on the E _m .	Riluzole, 1-EBIO, NS309, SKA-31	Charybdotoxin, TRAM-34, clotrimazole, ICA-17043, NS6180	73–79
	K _{Ca} 1.1	Large conductance calcium activated potassium channels/ BK _{Ca}	<i>KCNMA1</i> , <i>KCNMB1</i>	7	VSMC, platelets		NS1619, BMS-204352, flupirtine, cAMP, cGMP, carbon monoxide, epoxide of arachidonic acid, dehydrosoyasaponin-1, 17β-estradiol	TEA, charybdotoxin, iberiotoxin, Paxilline	2,3,61,74,80–87
K _v	K _v 1.2	Voltage-dependent (delayed rectifier) potassium channels	<i>KCNA2</i>	6	VSMC, EC	Induces VSMC relaxation by increasing K ⁺ efflux to cause hyperpolarization and thus, close calcium channels to prevent any further contractions.	Retigabine, NO, cAMP, cGMP	TEA, 4-AP, charybdotoxin, α-dendrotoxin, margatoxin, correolide	82,88,89
	K _v 1.3		<i>KCNA3</i>		VSMC, EC, macrophages, platelets, neutrophils			TEA, 4-AP, maurotoxin, α-dendrotoxin, margatoxin, Psora-4, PAP-1, correolide	79,82,88–95
	K _v 1.5		<i>KCNA5</i>		VSMC, EC			TEA, 4-AP, correolide	82,89,96
	K _v 2.1		<i>KCNB1</i>				Retigabine, NS1643	TEA, 4-AP, Ba ²⁺	65,97–99
	K _v 7.1		<i>KCNQ1</i>		VSMC, EC, cardiomyocytes, cells of the sinoatrial node		Flupirtine, R-L3, ML227, ML213, celecoxib, diclofenac	TEA, retigabine, Amiodarone, linopirdine, XE991	100–108
	K _v 7.4		<i>KCNQ4</i>		VSMC		Flupirtine, retigabine, ML227, ML213, BMS-204352, celecoxib, diclofenac	TEA, Linopirdine, XE991	100,102–107,109,110
K _{ATP}	K _{ir} 6.1	ATP-sensitive potassium channels	<i>KCNJ8</i>	2	VSMC, EC, cardiac myocytes, platelets, macrophages	Induces VSMC relaxation by increasing K ⁺ efflux to cause hyperpolarization and closure of calcium channels to prevent further contractions. Intracellular ATP inhibits K _{ATP} channels, resulting in a very low likelihood of these channels being open in most VSMC under normal conditions.	Diazoxide, pinacidil, levromacalim (lemacalim), aprikalim, norepinephrine, adenosine, prostacyclin forskolin, KMUP-1, nitroglycerine, cromakalim, nicorandil, minoxidil	Ba ²⁺ , Glibenclamide, Tolbutamide, propranolol†	62,111–121
	K _{ir} 6.2		<i>KCNJ11</i>						

(Continued)

Table 1. Continued

Fam-ily	Sub-type	Syn-onyms	Gene name (human)	No. of TM domains	Expressing vascular cells*	Functions	Agonists	Antagonists	References
K _{ir}	K _{ir} 2.1	Inward-rectifier potassium channels	KCNJ2	2	VSMC, cardiac myocytes, platelets, macrophages	Induces VSMC relaxation by allowing the selective flow of K ⁺ across the cell membrane primarily in the inward direction at negative E _m .	Zacopride, extracellular K ⁺	BaCl ₂ , 4-AP, TEA, Mg ²⁺ , CsCl, ML133, mepyramine, diphenhydramine	3,120,122–130
	K _{ir} 2.2		KCNJ12		VSMC, cardiac myocytes, EC				
K _{2p}	TWIK-2	Two-pore domain potassium channels	KCNK6	4	VSMC, EC	Provides a background potassium conductance that helps to maintain the resting E _m .	Arachidonic acid	Anandamide, Zn ²⁺	1,4,6,131,132

1-EBIO indicates 1-ethyl-2-benzimidazolinone; 4-AP, 4-aminopyridine; Ba²⁺, barium ion; BaCl₂, barium chloride; BK_{Ca}, large-conductance calcium-activated potassium channels; BMS-204352, flindokalner; Ca²⁺, calcium ion; CsCl, cesium chloride; EC, endothelial cells; E_m, membrane potential; ICA-17043, senicapoc; IK_{Ca}, intermediate-conductance calcium-activated potassium channels; K⁺, potassium ion; K_{2p}, 2-pore domain potassium channel; K_{ATP}, ATP-sensitive potassium channel; K_{Ca} 2.2, small-conductance calcium-activated potassium channel 2.2; K_{Ca} 2.3, small-conductance calcium-activated potassium channel 2.3; K_{ir}, inward-rectifier potassium channel; K_{ir} 6.1, inward-rectifier potassium channel 6.1; K_{ir} 6.2, inward-rectifier potassium channel 6.2; KMUP-1, (7-[2-[4-(2-chlorophenyl)piperazinyl]ethyl]-1,3-dimethylxanthine); K_v, voltage-gated potassium channel; K_v 1.2, voltage-gated potassium channel 1.2; K_v 1.3, voltage-gated potassium channel 1.3; K_v 1.5, voltage-gated potassium channel 1.5; K_v 7.1, voltage-gated potassium channel 7.1; K_v 7.4, voltage-gated potassium channel 7.4; Mg²⁺, magnesium ion; NS1619, [1,3-dihydro-1-[2-hydroxy-5-(trifluoromethyl)phenyl]-5-(trifluoromethyl)-2H-benzimidazol-2-one]; NS309, 6,7-dichloro-1H-indole-2,3-dione 3-oxime; NS6180, 4-[[3-(trifluoromethyl)phenyl]methyl]-2H-1,4-benzothiazin-3(4H)-one; PAP-1, 5-(4-phenoxybutoxy)psoralen; Psora-4, 4-(4-Phenylbutoxy)-7H-furo[3,2-g]benzopyran-7-one; R-L3, (3R)-5-(2-fluorophenyl)-3-(1H-indol-3-ylmethyl)-1-methyl-3H-1,4-benzodiazepin-2-one; SKA-31, naphtho[1,2-d]thiazol-2-ylamine; SK_{Ca}, small-conductance calcium-activated potassium channels; TEA, tetraethylammonium; TM, transmembrane; TRAM-34, triaryl methane-34; TWIK, tandem of P-domains in a weak inward rectifying potassium channel; UCL1684, ditrifluoroacetate hydrate; VSMC, vascular smooth muscle cells; XE991, 10,10-bis(4-Pyridinylmethyl)-9(10H)-anthracenone dihydrochloride; and Zn²⁺, zinc ion.

*Cells are listed in descending order based on their prominence
†Indirect inhibition.

peripheral artery disease, and stroke.^{10,144,145} This is due to the critical function of K⁺ channels in maintaining E_m, Ca²⁺ flux, and vasodilation.

In EC, impaired activity of SK_{Ca}, IK_{Ca}, and K_{ir} channels disrupts EDH and NO production, thereby compromising vasodilation and promoting endothelial dysfunction.^{146,147} Endothelial dysfunction increases oxidative stress and reactive oxygen species, while upregulating adhesion molecules like VE-cadherin (vascular endothelial cadherin) and proinflammatory cytokines, further decreasing NO bioavailability.^{148,149} As a result, EC further lose their vasoprotective function, leading to impaired vasodilation.

In response to endothelial dysfunction and environmental stimuli such as oxidative stress or cytokines, VSMC phenotype switching occurs, primarily regulated by IK_{Ca} and voltage-gated potassium channels encoded by the *KCNA3* gene channels (Figure 3).^{53–55} This switch is characterized by increased cell proliferation, migration into the intima, and excessive production of extracellular matrix components, such as matrix metalloproteins and degradation of collagen and elastin.¹⁵⁰ These changes contribute to aberrant vascular remodeling, vessel wall thickening, and stiffness, features that are commonly observed in atherosclerosis, aneurysm, and hypertension.^{151,152} Concurrently, altered function of BK_{Ca}, K_v, K_{ATP}, and K_{ir} leads to membrane depolarization, increased intracellular Ca²⁺, and sustained vasoconstriction.²⁰ Together, these interrelated processes occurring in EC and VSMCs disrupt vascular homeostasis, leading to elevated vascular resistance and impaired tissue perfusion. This in turn

accelerates pathological vascular remodeling and lays the groundwork for chronic vascular diseases such as hypertension, atherosclerosis, coronary artery disease, stroke, peripheral artery disease, and aortic aneurysm (Table 3). Hypertension will be discussed in the subsequent sections, given that the most evidence exists in this domain.

Cardiomyocytes

Familial genetic mutations, atrial remodeling, and altered K⁺ channel expression critically disrupt atrial repolarization, predisposing to atrial fibrillation (AF).^{15,163} Pathogenic variants identified in K_v1.5 (*KCNA5*), voltage-gated potassium channel encoded by the *KCND2* gene (K_v4.2), voltage-gated potassium channel encoded by the *KCND3* gene (K_v4.3), inward-rectifier potassium channels subunit 2.1 (*KCNJ2*), inward-rectifier potassium channels subunit 3.1/inward-rectifier potassium channels subunit 3.4 (*KCNJ3/KCNJ5*), and voltage-gated potassium channel encoded by the *KCNQ1* gene/voltage-gated potassium channel encoded by the *KCNH2* gene (K_v7.1/K_v11.1) dysregulate the corresponding repolarizing currents (ultra-rapid delayed rectifier potassium current [I_{Kur}], transient outward potassium current, and inward-rectifier potassium current).^{14,96,139,164–169} Loss-of-function mutations in K_v1.5 (*KCNA5*) and voltage-gated potassium channel encoded by the *KCND3* gene (K_v4.3), which reduce I_{Kur} and transient outward potassium current, respectively, lead to prolonged atrial APD, thereby increasing susceptibility to early afterdepolarizations and triggered activity.^{164,169}

Table 2. Atrial Potassium Ion Channels and Their Role in Action Potential During Cardiac Electrophysiology

Family	Type of K current	K channel subunit	Arrhythmias	References
K _{Ca}	I _{KCa}	SK1 (α)	Risk of lone AF	14,96,135
		SK2 (α)		
		SK3 (α)		
		SK4 (α)		
K _v	I _{tot}	K _v 4.2 (α)	Early-onset nocturnal paroxysmal AF	14,96,138
		K _v 4.3 (α)	Early-onset lone AF	13,14,96,139
	I _{Ks}	KCNE1 (β)	Early-onset lone AF	13,14,96
		KCNE2 (β)	AF	140
		KCNE5 (β)		141
	I _{Kr}	KCNE3 (β)	AF	13,96
	I _{Kur}	K _v 1.5 (α)	AF	13,96
		K _v β1 (β)		
K _{ATP}	I _{KATP}	SURA (β)	AF, Cantu syndrome	142
K _r	I _{Kach}	K _v 3.1 (α)	AF	96
		K _v 3.4 (α)	AF, SAN dysfunction	
K _{2P}	I _{KP}	K _{2P} 2.1 (α)	Cardiac conduction disorder	14,96
		K _{2P} 3.1 (α)		
		K _{2P} 17.1 (α)		143

AF indicates atrial fibrillation; Ca²⁺, calcium ion; I_{Kach}, acetylcholine-activated inward-rectifier potassium current; I_{KCa}, intermediate-conductance calcium-activated potassium channel; I_{Ks}, slow delayed rectifier potassium current; I_{tot}, voltage-gated potassium currents generating fast transient outward potassium current; I_{Kur}, ultrarapid delayed rectifier potassium current; K_{2P}, 2-pore-domain potassium channel; K_{2P}2.1, two-pore-domain potassium channel 2.1; K_{2P}3.1, two-pore-domain potassium channel 3.1; K_{2P}17.1, two-pore-domain potassium channel 17.1; K_{ATP}, ATP-sensitive potassium channel; K_{Ca}, Ca²⁺-activated potassium channel; KCNE, KCNE gene; K_v, inward-rectifier potassium channel; K_v3.1, inward-rectifier potassium channels subunit 3.1; K_v3.4, inward-rectifier potassium channels subunit 3.4; K_v, voltage-gated potassium channel; K_vβ1, voltage-gated potassium channel β-subunit 1; K_v1.5, voltage-gated potassium channel 1.5 encoded by the *KCNA5* gene; K_v4.2, voltage-gated potassium channel 4.2 encoded by the *KCND2* gene; K_v4.3, voltage-gated potassium channel 4.3 encoded by the *KCND3* gene; SAN, sinoatrial node; SK, small-conductance calcium-activated potassium channel; and SURA, sulfonyleurea receptor A.

In contrast, gain-of-function mutations in voltage-gated potassium channel encoded by the *KCNQ1* gene (K_v7.1) and voltage-gated potassium channel encoded by the *KCNH2* gene (K_v11.1), which enhance I_{Ks} and I_{Kr} currents, shorten APD and refractory periods, facilitating reentry mechanisms that sustain AF.^{164,169} In addition, dysregulation of SK channels encoded by *KCNN3* (SK3) further modulates atrial excitability, indicating that both genetic and remodeling-induced alterations of K⁺ channel function are central to the electrophysiological basis of AF.¹⁴

ROLE OF K⁺ CHANNELS IN HYPERTENSION

In the context of hypertension, dysfunctional potassium channels may promote depolarization, resulting in

a more positive E_m, causing vasoconstriction (Figure 4). Long-term increase in peripheral resistance is a key factor in the development of hypertension.¹⁸ In the heart, potassium channels are responsible for repolarization of the myocytes, which is necessary for heart rhythm and contractility.¹⁶ Another vital feature in hypertension is the regulation of Na⁺ resorption via sodium-potassium ATPase channels in tubular cells of the kidney. Consequently, dysfunction in potassium channel activity can disrupt the balance of Na⁺ and K⁺ levels, leading to increased Na⁺ reabsorption and fluid retention. This imbalance could lead to hypertension.¹⁷⁰ Decreased renal blood pressure and reduced renal tubular Na⁺ content activate the juxtaglomerular apparatus in the kidney and lead to cleavage of prorenin to renin. Renin converts angiotensinogen, a liver-produced protein, into angiotensin I (Figure 4). Angiotensin I is then converted to angiotensin II by angiotensin II-converting enzyme in the lungs.¹⁷¹ Angiotensin II inhibits K_{ir} and K_{2P} channels in zona glomerulosa cells of the adrenal cortex.¹⁷² Upon inhibition of these potassium channels, zona glomerulosa cell membranes depolarize and activate VGCC, causing increased cytosolic Ca²⁺ concentration.¹⁷³ This rise in Ca²⁺ levels stimulates the expression of CYP11B2 (cytochrome P450 family 11 subfamily B member 2) in mitochondria to synthesize aldosterone.¹⁷⁴ Aldosterone plays a key role in regulating blood pressure by increasing Na⁺ reabsorption and K⁺ excretion via the sodium-potassium ATPase pumps of the distal convoluted tubule in the kidneys. These pumps create a concentration gradient that favors Na⁺ reabsorption from the urine by actively transporting Na⁺ out of the renal tubular cells into the bloodstream in exchange for K⁺ transported into the cells. This concentration gradient passively transports water due to osmotic forces, leading to an increase in blood volume and consequently elevated blood pressure.¹⁷¹

Evidence from Animal Studies

Large-Conductance Calcium-Activated K⁺ Channels

Hypertension has been shown to both enhance and reduce the expression and function of BK_{Ca} channels. Several rodent studies have reported increased expression and activity of BK_{Ca} channels in hypertension compared with nonhypertensive controls (Table 4). Studies involving Sprague-Dawley rats with renovascular hypertension induced via the 2-kidney-1-clip method found that BK_{Ca} channel protein expression was significantly higher in mesenteric arteries compared with sham controls.⁸⁰ The increased BK_{Ca} channel protein levels were more significant in adult spontaneously hypertensive rats (SHR) than both the younger counterparts and nonhypertensive Wistar-Kyoto rats (WKY) controls.⁸¹ Similarly, cerebral arteries of SHR exhibited 2 to 4 times greater constriction in response to the BK_{Ca} inhibitor iberiotoxin, associated with a 4.7-fold

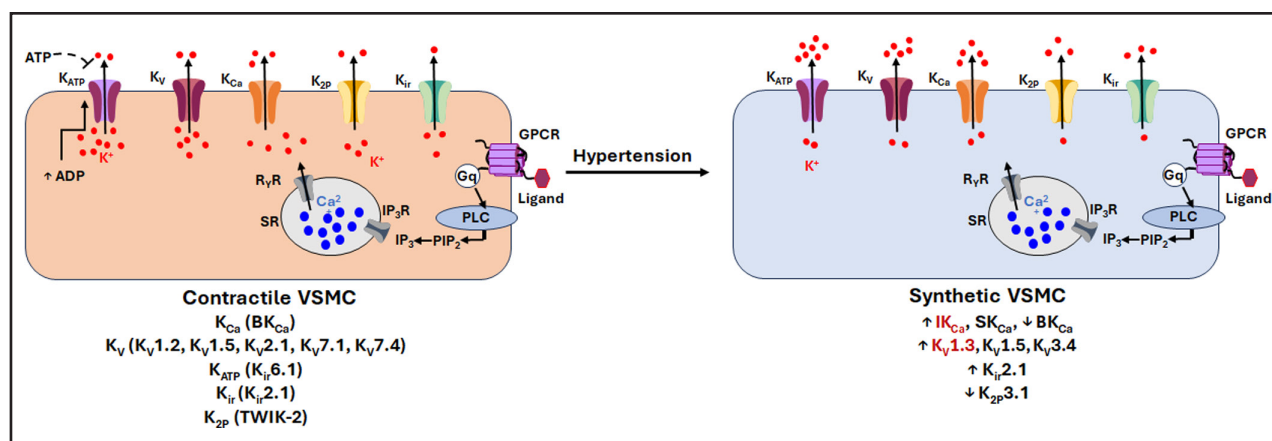


Figure 3. Potassium channel expression changes during vascular smooth muscle cell phenotype switching in response to hypertension.

Contractile vascular smooth muscle cells (VSMC) express large-conductance calcium-activated potassium channel (BK_{Ca}), voltage-gated potassium channel (K_V), ATP-sensitive potassium channel (K_{ATP}), inward-rectifier potassium channel (K_{ir}), and 2-pore domain potassium channel (K_{2P}), which maintain membrane hyperpolarization and regulate vascular tone. Under hypertensive conditions, dysregulation of intermediate-conductance calcium-activated potassium channel encoded by the *KCNA3* gene ($K_V1.3$), voltage-gated potassium channel encoded by the *KCNA5* gene ($K_V1.5$), $K_V3.4$, inward-rectifier potassium channels subunit 2.1 ($K_{ir}2.1$), and TWIK (tandem of P-domains in a weak inward rectifying potassium channel)—related acid-sensitive potassium channel 1, a member of the 2-pore domain potassium ($K_{2P}3.1$) channels promotes phenotypic switching of VSMC from a contractile to a synthetic state, enhancing cellular proliferation and migration. Potassium channel subtypes that primarily regulate VSMC phenotypic switching are highlighted in red. Ca^{2+} indicates calcium ion; ER, endoplasmic reticulum; GPCR, G-protein-coupled receptors; Gq, G-protein q; IP₃, inositol 1,4,5-trisphosphate; IP₃R, IP₃ receptor; K^+ , potassium ion; K_{Ca} , Ca^{2+} -activated potassium channel; $K_{ir}6.1$, inward-rectifier potassium channels subunit 6.1; $K_V1.2$, voltage-gated potassium channel 1.2; $K_V2.1$, voltage-gated potassium channel 2.1; $K_V3.4$, voltage-gated potassium channel 3.4; $K_V7.1$, voltage-gated potassium channel 7.1 encoded by the *KCNQ1* gene; $K_V7.4$, voltage-gated potassium channel 7.4; NOS, NO synthase; PIP₂, phosphatidylinositol-bisphosphate; PLC, phospholipase C; RyR, ryanodine receptor; SK_{Ca} , small-conductance calcium-activated potassium channel; and SR, sarcoplasmic reticulum.

increase in BK_{Ca} channel density and a 4.1-fold higher expression of the BK_{Ca} α -subunit compared with WKY controls.¹⁷⁵ In a study investigating the effect of aging on arterial tone, BK_{Ca} current and channel expression were measured in VSMC from SHR of different age groups. Results showed that BK_{Ca} α -subunit expression increased with age, particularly in the cytoplasm of SHR, and that BK_{Ca} current density rose in aging SHR but remained unchanged in WKY rats.¹⁷⁶ These findings suggest that BK_{Ca} channels are upregulated in response to hypertension, potentially serving a compensatory role by stimulating vasodilation to counteract elevated blood pressure, in some cases, secondary to aging. Interestingly, 8 weeks of aerobic exercise in SHR led to a significant reduction in the expression of the BK_{Ca} β 1-subunit, while the α -subunit levels remained unchanged.⁸² These studies indicate that exercise could modify or potentially restore BK_{Ca} channel function and improve vascular health in the hypertensive state. This aligns with the upregulation of BK_{Ca} channels observed in response to hypertension, supporting the notion that these channels are part of a compensatory mechanism. In contrast, hypertension induced by the continuous release of angiotensin II through osmotic minipumps in Sprague-Dawley rats caused a decrease in the expression of the β 1-subunit compared with the α -subunit in cerebral arteries.¹⁷⁷ This reduction in β 1-subunit expression was linked to a decreased Ca^{2+} sensitivity of the

BK_{Ca} channels, leading to impaired BK_{Ca} channel function. A decreased expression of β 1-subunit was identified in vessels from SHR, as well as in WKY rats, whose blood pressure levels were intermediate between those of Sprague-Dawley rats and SHR.¹⁸⁶ The differences in BK_{Ca} channel expression and function may be due to variations in methodology, vascular bed heterogeneity, and strain- or species-specific factors.

In stroke-prone SHR, contractile responses to potassium channel antagonists like tetraethylammonium and barium chloride were markedly greater compared with control rats.¹⁷⁸ Responses to other antagonists such as 3,4-aminopyridine and sparteine did not differ, however, indicating that while BK_{Ca} channel activity is altered, its regulation involves complex interactions with other potassium channels, especially in the presence of other cardiovascular comorbidities. In this context, interventions such as exercise and alterations in subunit expression could offer potential strategies for modulating BK_{Ca} channel function and improving hypertension management. In a BK_{Ca} β 1 knockout mouse model, blood pressures measured via the tail cuff method were shown to be elevated as a compensatory response to normalize vascular tone via restoring Ca^{2+} transient outward current coupling. The researchers also found cerebral arteries from β 1 subunit knockout mice were more constricted, and the animals had substantially higher resting mean arterial blood pressure than controls.^{179,180} However, these results were not

Table 3. Summary of Potassium Ion Channels Associated With Vascular Diseases

Disease	Potassium ion channel involved	Associated mechanisms
Hypertension ^{20,146,153}	↓ Endothelial IK _{Ca} , SK _{Ca} channels causing endothelial dysfunction	Sustained depolarization, increased vascular resistance, and tone
	↓ Vascular BK _{Ca} and K _v impair vasodilation	
Atherosclerosis ^{154,155}	↓ Endothelial IK _{Ca} , SK _{Ca} channels causing endothelial dysfunction	Increased vascular resistance promotes plaque formation and narrowing of vessel that affects vascular tone
	↓ Vascular BK _{Ca} and K _{ATP} impair vasodilation	
	↑ K _v cause VSMC proliferation	
Coronary artery disease ^{72,156–158}	↓ Endothelial IK _{Ca} , SK _{Ca} channels causing endothelial dysfunction	Myocardial ischemia and infarction, promote plaque growth and stability
	↓ Vascular K _{ATP} and K _{ir} impair vasodilation	
	↑ K _v cause VSMC proliferation	
Stroke ^{159,160}	↓ Endothelial IK _{Ca} , SK _{Ca} channels causing endothelial dysfunction	Chronic hypoperfusion
	↓ Vascular BK _{Ca} and K _{ATP} impair vasodilation	
Peripheral artery disease ^{153,155,157}	↓ Endothelial IK _{Ca} , SK _{Ca} channels causing endothelial dysfunction	Impaired microvascular perfusion, increased vasoconstriction and neointimal hyperplasia, and promotes plaque growth
	↓ Vascular BK _{Ca} and K _{ATP} impair vasodilation	
	↑ K _v cause VSMC proliferation	
Aortic and peripheral artery aneurysm ^{161,162}	↓ Endothelial IK _{Ca} , SK _{Ca} channels causing endothelial dysfunction	Sustained depolarization, increased vascular resistance, promotes aneurysm formation through VSMC phenotypic switching and extracellular matrix degradation.
	↓ Vascular BK _{Ca} , K _v and K _{ATP} impair vasodilation	
	↑ K _v cause VSMC proliferation	

BK_{Ca} indicates large-conductance calcium-activated potassium channel; Ca²⁺, calcium ion; IK_{Ca}, intermediate-conductance calcium-activated potassium channel; K_{ATP}, ATP-sensitive potassium channel; K_{Ca}, Ca²⁺-activated potassium channel; K_{ir}, inward-rectifier potassium channel; K_v, voltage-gated potassium channel; SK_{Ca}, small-conductance calcium-activated potassium channel; and VSMC, vascular smooth muscle cell.

reproduced when blood pressure was measured using telemetry.¹⁸¹ Further well-designed studies with validated measurement methods are needed.

Voltage-Gated K⁺ Channels

The impact of hypertension on K_v channel expression and function remains uncertain.^{187–189} Electrophysiological studies in VSMCs from hypertensive animal models have reported heterogeneous findings, including increased K_v current density in mesenteric^{88,190} and tail⁸⁸ arteries from SHR, reduced K_v channel expression and function in mesenteric artery,^{191,192} thoracic aorta,^{82,182} and interlobar artery¹⁹³ of hypertensive rats, and no significant changes in K_v current density in other vascular

beds, such as the aorta of stroke-prone SHR.^{194,195} In contrast to the generally enhanced activity of BK_{Ca} channels observed in hypertension, most evidence from rodent models, particularly comparison between SHR and normotensive WKY rat controls, indicates a decreased function and expression of arterial K_v channels (Table 4). Electrophysiological analysis revealed significantly reduced K_v currents in SHR thoracic aorta compared with WKY, suggesting a potential contribution of these channels towards resting K⁺ conductance under normotensive conditions compared with hypertension.¹⁸² Li et al⁸² observed a significantly lower protein expression levels of K_v1.2 and K_v1.5 channels in the thoracic aorta of SHR compared with WKY rats, which was reversed by aerobic exercise training, suggesting that exercise mitigates hypertension-induced changes in K_v channel expression. Similarly, Zhang et al found reduced K_v1.2 and K_v1.5 expression levels in mesenteric arterial smooth muscle cells of SHR compared with WKY rats, which was likely due to altered potassium channel composition.⁸¹ Another study reported diminished K_v7.4 channel protein expression and function in the thoracic aorta and mesenteric arteries of SHR and angiotensin-induced hypertensive mice when compared with normotensive controls.¹⁹⁶ Similarly, Zavaritskaya et al¹⁹⁷ reported reduced K_v7.4 protein expression and function in mesenteric arteries from SHR. However, in the gracilis artery, K_v7.4 expression and function appeared to remain unaffected in SHR.¹⁹⁷

A study comparing gene expression of K_v channels from the Kv1.X family across various tissues (heart, aorta, tail artery, and mesenteric artery) found that K_v1.2 and K_v1.5 were the most highly expressed, with K_v1.2 levels significantly higher in all SHR tissues and higher K_v1.5 levels in SHR hearts compared with WKY controls. Moderate levels of voltage-gated potassium channels encoded by the *KCNA3* gene and K_vβ1.1 were also observed in all tissues, both higher in SHR than in WKY rats. This differential expression was suggested to be associated with increased potassium channel current density and altered voltage-dependent activation in SHR.⁸⁸ The differential expression also varied between protein and mRNA levels. For instance, Carr et al¹⁸³ demonstrated that protein levels of K_v7.4, but not the mRNA (*Kcnq4*), were significantly reduced in renal and mesenteric arteries of SHR compared with normotensive Wistar rats, possibly due to posttranslational modification of the protein. These findings suggest that reduced vascular K_v channel activity in hypertensive rodent models contributes to depolarized E_m, increased intracellular Ca²⁺ levels and enhanced vascular tone observed in chronic hypertension. The variations observed in K_v current density and channel expression could be attributed to differences in experimental conditions, the different rodent models of hypertension used and the regional variations in the blood vessels studied.

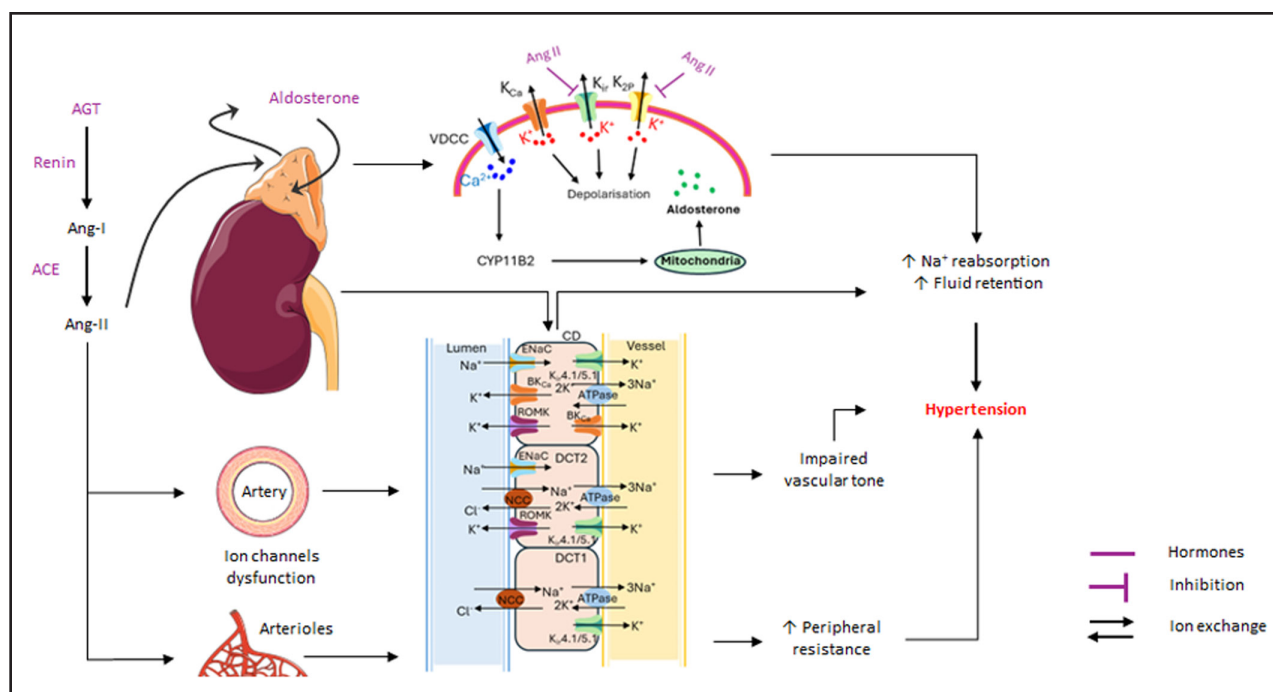


Figure 4. Schematic representation of the role of potassium ion (K⁺) channels in hypertension.

This figure depicts how activation of the renin-angiotensin-aldosterone system (RAAS) and subsequent dysregulation of vascular K⁺ channels lead to hypertension. Activation of RAAS increases renal sodium ion (Na⁺) reabsorption and fluid retention, leading to intravascular volume expansion. Concurrently, impaired vascular K⁺ channel activity disrupts arterial tone and enhances vasoconstriction, thereby increasing peripheral vascular resistance and blood pressure. ACE indicates angiotensin converting enzyme; AGT, angiotensinogen; Ang I, angiotensin I; Ang II, angiotensin II; BK_{Ca}, large-conductance calcium-activated potassium channel; Ca²⁺, calcium ion; Cl⁻, chloride ion; CD, collecting duct; CYP11B2, cytochrome P450 family 11 subfamily B member 2; DCT1, early distal convoluted tubule; DCT2, late distal convoluted tubule; ENaC, epithelial sodium channel; K_{2p}, 2-pore domain potassium channel; K_{Ca}, calcium-activated potassium channel; K_{ir}/K_{ir} 4.1/5.1, inward-rectifier potassium channel; NCC, sodium chloride cotransporter; ROMK, renal outer medullary potassium channel; and VDCC, voltage-dependent Ca²⁺ channel.

ATP-Sensitive and Inward-Rectifier K⁺ Channels

A few studies highlighted that vascular K_{ATP} channel function is compromised during hypertension (Table 4). One study examined basilar artery dilation in response to aprikalim, a direct K_{ATP} channel activator, which was significantly reduced in stroke-prone SHR compared with WKY rats, indicating impaired K_{ATP} function in this model.¹¹¹ Another study investigating genetically hypertensive mice showed a reduced mRNA expression level of the genes encoding K_{ir}2.1, K_{ir}4.1, 6.X subunit of ATP-sensitive potassium channel and SUR2 (sulfonylurea receptor subunit 2) compared with normotensive controls. In addition, there was a decrease in activity of both K_{ir} and K_{ATP} channels compared with normotensive mice, leading to a more depolarized resting E_m and increased vascular reactivity, thus suggesting a pivotal role of K_{ATP} channels in the vascular remodeling in hypertension.¹⁸⁴ Similarly, vasodilation triggered by K_{ATP} channel openers is diminished in the brains of stroke-prone SHR¹⁹⁸ and in the kidneys of SHR compared with that in WKY rats.¹⁹⁹ Blanco-Rivero et al²⁰⁰ reported a significant decrease in mRNA and protein expression for K_{ir}6.1 and SUR2 in vessels from SHR compared with WKY controls, but patch clamp experiments showed no reduction in K_{ATP}

channel currents and no impact on pinacidil-induced relaxation. To better understand the association between hypertension and K_{ATP} channel expression and function, further evidence is needed.

Reduced activity of K_{ir} channels was confirmed in the deep femoral artery and cerebral arteries in rats with angiotensin II-induced hypertension compared with sham-operated normotensive controls, along with an impaired K⁺ induced vasodilation in the deep femoral artery.¹⁸⁵ Furthermore, 2 weeks of exercise training reversed the impact of hypertension on K_{ir} channel currents and potassium-induced dilation in this artery. In cerebrovascular myocytes, however, although exercise restored K_{ir} channel currents, potassium-induced dilation remained impaired.¹⁸⁵ This evidence indicates regional differences in the response to exercise. Earlier studies have demonstrated that exercise training enhances both the expression and function of K_{ir} channels in VSMCs.²⁰¹ The authors concluded that decreased K⁺ conductance was associated with increased peripheral arterial resistance in hypertension, while the restoration of K⁺ conductance with exercise training suggested these channels to have potential antihypertensive effects. Tajada et al¹⁸⁴ reported that reduced inward-rectifier potassium

Table 4. Animal Studies Reporting the Relative Expression Changes and Activity of Potassium Ion Channels in Hypertension

Family	Subtype	Age/sex/strain/species	Model	Tissue used	Groups (n)	Relative expression	Channel activity	References
K _{Ca}	BK _{Ca} α	Adult/male/Sprague-Dawley rats	Hypertensive model-2-kidney-1-clip method	Mesenteric artery	Sham (6)	0.47±0.03	13.8±0.4*	80
					CIHH (6)	0.49±0.04	21.1±1.0*	
					RVH (6)	0.49±0.03	8.0±0.2*	
					RVH + CIHH (6)	0.49±0.01	10.6±0.4*	
	BK _{Ca} β1				Sham (6)	0.19±0.01	13.8±0.4*	
					CIHH (6)	0.31±0.03	21.1±1*	
					RVH (6)	0.08±0.01	8.0±0.2*	
					RVH + CIHH (6)	0.14±0.01	10.6±0.4*	
BK _{Ca}	12–16-wk-old/male wk/SHR and WKY rat	SHR	Cerebral artery	SHR (27)	5.31±1.69	0.6±0.1*	175	
				WKY (27)	21.97±6.43	2.8±0.5*		
BK _{Ca} α	12-wk-old/ male /SHR and WKY rat	SHR	Thoracic aorta	WKY (12)	0.97±0.01	24.3±2.5	82	
				SHR-SED (12)	1.04±0.14	62.5±5.8		
				SHR-EX (12)	0.90±0.09	48.1±3.9		
BK _{Ca} β1	WKY (12)	0.97±0.01	24.3±2.5					
	SHR-SED (12)	4.19±0.79	62.5±5.8					
	SHR-EX (12)	2.29±0.42	48.1±3.9					
BK _{Ca} α	4–6-wk, 16–18-wk and 40–42-wk-old/ male /SHR and WKY rats	SHR	Mesenteric artery	Very young SHR (20)	0.38±0.06†	18.4±2.6‡	176	
				Young SHR (20)	0.43±0.01†	18.1±2.0‡		
				Adult SHR (20)	0.43±0.03†	18.5±2.0‡		
				Very young WKY (20)	0.39±0.03†	18.0±2.7‡		
				Young WKY (20)	0.41±0.01†	18.3±2.3‡		
				Adult WKY (20)	0.44±0.02†	19.7±3.7‡		
BK _{Ca} β1	Very young SHR (20)	0.39±0.05†	18.4±2.6‡					
	Young SHR (20)	0.85±0.02†	18.1±2.0‡					
	Adult SHR (20)	0.85±0.03†	18.5±2.0‡					
	Very young WKY (20)	0.39±0.03†	18.0±2.7‡					
	Young WKY (20)	0.42±0.01†	18.3±2.3‡					
	Adult WKY (20)	0.45±0.02†	19.7±3.7‡					
BK _{Ca} α	Male/Sprague-Dawley rats	Subcutaneous implantation of Ang osmotic minipump	Cerebral artery	HT (9)	0.06±0.01†	43.2±9.2§	177	
				NT (8)	0.07±0.01†	112.4±26.8§		
BK _{Ca} β1				HT (9)	0.02±0.01†	43.2±9.2§		
				NT (8)	0.07±0.01†	112.4±26.8§		
BK _{Ca}	16–20-wk-old/ male and female SP-SHR and WKY rats	SP-SHR	Carotid artery	SP-SHR–TEA 0.1 mmol/L (control; 24)	NR	177.3±30.2	178	
				WKY–TEA 0.1 mmol/L (control; 24)	NR	251.3±23.4		
				SP-SHR–TEA 0.1 mmol/L (Ca-free; 24)	NR	4.9±0.9		
				WKY–TEA 0.1 mmol/L (Ca-free; 24)	NR	12.7±5.8		
				SP-SHR–BaCl ₂ 1 mmol/L (control; 24)	NR	81.8±3.9		
				WKY–BaCl ₂ 1 mmol/L (control; 24)	NR	127.3±25.3		
				SP-SHR–BaCl ₂ 1 mmol/L (Ca-free; 24)	NR	32.5±7.2		
				WKY–BaCl ₂ 1 mmol/L (Ca-free; 24)	NR	52.6±11.7		

(Continued)

Table 4. Continued

Family	Subtype	Age/sex/strain/species	Model	Tissue used	Groups (n)	Relative expression	Channel activity	References
	BK _{Ca}	6–8-month-old and pre-hypertensive 3–4 wk-old/male/SHR and WKY rats	SHR	Mesenteric artery	SHR young (8)	0.72±0.02	NR	81
					Adult SHR (8)	1.25±0.06	NR	
					WKY young (8)	0.77±0.02	NR	
					Adult WKY (8)	0.86±0.01	NR	
	BK _{Ca} β1	3–8-mo-old/male/129/svjxC57BL mice	<i>Kcnmb1</i> ^{−/−}	Cerebral artery	Control (12)	NR	37.8±1.6§	179
					Knockout (12)	NR	6.7±10.5§	
	BK _{Ca} β1	Male/129/svjxC57BL/6J cross	<i>Kcnmb1</i> ^{−/−}	Thoracic aorta	Control (25)	NR	3.2±0.3	180
					Knockout (31)	NR	5.1±0.6	
	BK _{Ca} β1	10–12 wk/ male / C57BL/6 mice	<i>Kcnmb1</i> ^{−/−}	Mesenteric artery	Control (14)	NR	6.3±0.4	181
					Knockout (14)	NR	7.2±0.3	
K _v	K _v 1.2	12-wk-old/ male /SHR and WKY rat	SHR	Thoracic aorta	Control (WKY; 12)	0.96±0.01	52.5±3.2	82
					SHR-SED (12)	0.74±0.07	22.5±2.9	
					SHR-EX (12)	0.85±0.01	44.9±1.4	
					Control (WKY; 12)	0.96±0.01	52.5±3.2	
					SHR-SED (12)	0.37±0.02	22.5±2.9	
					SHR-EX; 12)	1.11±0.13	44.9±1.4	
	K _v 1.2	12–15-wk-old/male/ SHR and WKY rats	SHR	Heart	WKY (10)	4.62±0.44†	NR	88
					SHR (10)	9.69±2.10†	NR	
				Aorta	WKY (10)	3.54±0.74†	NR	
					SHR (10)	6.92±1.69†	NR	
				Tail artery	WKY (10)	2.16±0.46†	NR	
					SHR (10)	8.00±0.76†	NR	
				Mesenteric artery	WKY (10)	13.54±0.92†	7.3±0.8*	
					SHR (10)	21.23±2.46†	10.9±1.2*	
	K _v 1.2	6–8-month-old and prehypertensive 3–4 wk old/male/SHR and WKY rats	SHR	Mesenteric artery	SHR young (8)	0.81±0.02	NR	81
					Adult SHR (8)	0.93±0.09	NR	
					WKY young (8)	0.87±0.03	NR	
					Adult WKY (8)	1.18±0.01	NR	
					SHR young (8)	0.90±0.02	NR	
					Adult SHR (8)	0.84±0.09	NR	
					WKY young (8)	0.90±0.03	NR	
					Adult WKY (8)	1.15±0.04	NR	
	K _v	12-wk-old/male/SHR and WKY rats	SHR	Thoracic aorta	SHR (16)	NR	3.7±0.2*	182
					WKY (18)	NR	6.1±0.6*	
	K _v 7.4 (KCNC4)	12-wk-old/male/SHR and WKY rats	SHR	Thoracic aorta	WKY (NT; 6)	1.42±0.37	NR	183
					SHR (6)	0.56±0.13	NR	
				Mesenteric artery	WKY (NT; 6)	0.58±0.14	NR	
					SHR (6)	0.46±0.13	NR	
				Renal artery	WKY (NT; 6)	1.05±0.22	NR	
					SHR (6)	0.55±0.18	NR	
				Middle cerebral artery	WKY (NT; 6)	0.94±0.24	NR	
					SHR (6)	1.00±0.26	NR	
K _{ATP}	K _{ATP}	6–9-month-old/male/SP-SHR and WKY rats	SP-SHR	Basilar artery	WKY– (Aprikalim; 15)	NR†	31.0±5.0	111
					SP-SHR– (Aprikalim; 15)	NR	5.0±1.0	
	K _{ir} 6.x	Normotensive BPN/3 and hypertensive BPH/2 mice	Hypertensive mice	Mesenteric artery	Normotensive (BPN; 15)	13.31±0.81	9.5±0.5*	184
					Hypertensive (BPH; 15)	0.89±5.00	3.1±0.2*	

(Continued)

Table 4. Continued

Family	Subtype	Age/sex/strain/species	Model	Tissue used	Groups (n)	Relative expression	Channel activity	References
K _{ir}	K _{ir} 2.1	Normotensive BPN/3 and hypertensive BPH/2 mice	Hypertensive mice	Mesenteric artery	Normotensive (BPN; 15)	1.53±0.21	11.5±0.8*	184
					Hypertensive (BPH; 15)	1.15±0.09	4.8±0.4*	
					Normotensive (BPN; 15)	1.68±0.05	11.5±0.8*	
					Hypertensive (BPH; 15)	0.96±0.04	4.8±0.4*	
	K _{ir}	8-wk-old/male/Sprague-Dawley rats	Ang-induced hypertensive rats (Ang-HT)	Femoral artery	Sham (72)	NR	7.4±0.5*	185
					Sham-ET (16)	NR	9.5±0.4*	
					Ang-HT (59)	NR	5.6±0.4*	
					Ang-HT rats that underwent regular exercise training (HT-ET; 21)	NR	10.1±1.1*	
				Cerebral artery	Sham (72)	NR	10.9±2.2*	
					Sham-ET (16)	NR	10.5±2.7*	
					Ag-HT (59)	NR	3.6±0.7*	
					Ang-HT rats that underwent regular exercise training (HT-ET; 21)	NR	3.2±0.5*	

Protein expression levels were reported relative to GAPDH or β -actin using Western blotting (unless otherwise specified). Electrical activity in the tissue organ bath was reported as vascular reactivity in terms of tension in milligrams (mg) or millinewtons (mN) unless otherwise specified. Ang indicates angiotensin II; BaCl₂, barium chloride; BK_{Ca}, large-conductance calcium-activated potassium channels; BPH, hypertensive mice; BPN, normotensive mice; Ca²⁺, calcium ion; CIHH, chronic intermittent hypobaric hypoxia; EX/ET, exercise training group; HT, hypertensive; K_{ATP}, ATP-sensitive potassium channel; Kcnmb^{-/-}, β 1-knockout; K_{ir}, inward-rectifier potassium channel; K_{ir}2.1, inward-rectifier potassium channel 2.1; K_{ir}4.1, inward-rectifier potassium channel 4.1; K_{ir}6.x, 6.X subunit of ATP-sensitive potassium channel; K_v, voltage-gated potassium channel; K_v1.2, voltage-gated potassium channel 1.2; K_v1.5, voltage-gated potassium channel 1.5 encoded by the *KCNA5* gene; NR, not reported; NT, normotensive; pA/pF, picoamperes/picofarads; RVH, renovascular hypertension; SED, sedentary group; SHR, spontaneously hypertensive rat; SP-SHR, stroke-prone spontaneously hypertensive rat; TEA, tetraethylammonium; and WKY, Wistar-Kyoto rat.

*Electrical activity in the patch clamp measured using whole-cell current (pA/pF).

†Gene expression levels were reported relative to GAPDH or β -actin using reverse transcriptase quantitative polymerase chain reaction.

‡Electrical activity in the patch clamp measured using membrane capacitance (pF).

§Electrical activity in the patch clamp measured using channel conductance (BK_{Ca} current/Ca²⁺ sparks amplitude).

||Percentage change of diameter.

channels subunit 2.1 expression from hypertensive mice compared with normotensive controls. This reduction in expression was linked to a decrease in K_{ir} channel current density, VSMC depolarization, and increased myogenic reactivity of resistance arteries in hypertensive mice, which may contribute to the hypertension observed in this model. However, despite the reduced K_{ir} currents, barium ion-induced constriction of mesenteric arteries was not diminished in vessels from hypertensive mice.¹⁸⁴ These findings suggest that, even with reduced expression and currents, K_{ir} channel function was preserved, likely due to remodeling of other ion channels in the VSMC of hypertensive mice to maintain homeostasis.¹⁸⁴

Downregulation of K_v channels and upregulation of BK_{Ca} channels reflect interconnected feedback to preserve E_m during depolarization. For instance, K_v channel suppression reduces repolarizing outward K⁺ currents, promoting depolarization and VGCC activation, which can trigger compensatory BK_{Ca} upregulation via Ca²⁺ dependent pathways, enhancing hyperpolarization as a counterbalance to limit excessive contraction in resistance arteries.^{202,203}

Evidence From Human Studies

Two studies investigated BK_{Ca} activity and expression levels in freshly isolated human mesenteric artery smooth

muscle cells from 110 hypertensive and 164 nonhypertensive cancer patients undergoing surgery (Table 5).^{204,205}

One study categorized participants into 3 different groups based on age and hypertension status: younger patients without hypertension, older patients without hypertension, and older patients with hypertension.²⁰⁵ Both studies found that whole-cell current density, spontaneous transient outward currents, and Ca²⁺ sensitivity in the artery smooth muscle cells were significantly reduced in older patients with hypertension compared with patients with normotension.^{204,205} However, a limitation of the first study is that it did not evaluate the alteration of BK_{Ca} channel function in young patients with hypertension.²⁰⁵ In contrast to the downregulation observed in human studies, 6 out of 7 studies of rodent models of hypertension reported upregulation of both BK_{Ca} current and expression levels.^{80–82,175,176,178} There are several possible explanations for the different findings in rodent models and human samples. First, human studies included patients who were aged between 40 and 73 years,²⁰⁵ whereas rodent studies used animals only 1 to 8 months old^{182,175,176,179} that would be comparable to young human adults. In addition, the duration of hypertension tested in rodent models ranged from 1 to 8 weeks,^{82,177} whereas hypertension in humans manifests over multiple years.^{204,205} Furthermore, the tissues examined in rodent and human studies varied

Table 5. Human Studies Reporting the Relative Expression Changes and Activity of Potassium Ion Channels in Hypertension

Family	Subtype	Location	Condition/ cells studied	Group, n	Age, y (mean±SD)	Male sex, %	Relative gene expression	Relative protein expression	Channel activity (pA)	References
K _{ca}	BK _{Ca} α	Mes- enteric artery	SMC from patients with hypertension undergoing gastrointestinal surgeries	Patients with normotension (83)	56.0±10.4	72.3	0.012±0.002	0.43±0.17	23.5±2.3	204
				Patients with hypertension (66)	59.1±10.1	68.2	0.008±0.002	0.47±0.11	16.7±2.3	
	BK _{Ca} β			Patients with normotension (83)	56.0±10.4	72.3	0.021±0.002	2.45±0.47	23.5±2.3	
				Patients with hypertension (66)	59.1±10.1	68.2	0.012±0.005	1.22±0.22	16.7±2.3	
	BK _{Ca} α	Mes- enteric artery	SMC from patients with hypertension with cancer	YNMP (29)	40.0±1.0	62.1	1.000±0.120	0.57±0.08	32.9±1.8	205
				ONMP (52)	70.6±0.7	75.0	0.920±0.200	0.55±0.08	31.6±4.4	
	OHPP (44)			73.0±0.9	68.2	0.610±0.140	0.22±0.06	30.3±1.6		
	BK _{Ca} β			YNMP (29)	40.0±1.0	62.1	1.040±0.190	2.17±0.38	32.9±1.8	
				ONMP (52)	70.6±0.7	75.0	0.752±0.080	1.87±0.27	31.6±4.4	
				OHPP (44)	73.0±0.9	68.2	0.420±0.010	1.03±0.07	30.3±1.6	

Protein expression and gene expression levels were reported relative to β-actin using Western blotting and reverse transcriptase quantitative polymerase chain reaction, respectively. Electrical activity in the patch clamp was measured using whole-cell current. BK_{Ca} indicates large-conductance calcium-activated potassium channels; n, sample number, OHPP, older hypertensive patient, ONMP, older normotensive patient, pA, picoamperes, SMC, smooth muscle cells, and YNMP, young normotensive patient.

substantially. In the human investigations, BK_{Ca} channels were studied in mesenteric arteries,^{204,205} whereas in the rodent experiments BK_{Ca} channels were tested in samples from the thoracic aorta,^{82,180} and mesenteric,^{80,81,176} cerebral,^{175,177,179} and carotid¹⁷⁸ arteries. Importantly, it should also be noted that the rodent models studied lack comorbidities and exposure to multiple different medications used to treat hypertension, unlike patients. Also, frequently, there are health problems other than hypertension that lead to patients' providing samples for testing. For example, Cheng et al²⁰⁵ reported that patients with hypertension presented with gastrointestinal tumors, polyps, and intestinal obstruction, leading to them having laparotomies during which the mesenteric arteries were obtained for testing. The influence of comorbidities and multiple medications in patients may mean that the findings are not useful for judging the role of the ion channels in hypertension, which are better judged from findings in the more controlled experimental setting of rodent model studies. On the contrary, it is possible that the rodent experiments do not translate to patients. Which of these theories is correct is currently unclear.

Inherited Factors

Mendelian Randomization Studies

Somatic mutations of potassium channels were observed in patients with hypertension with an aldosterone-producing adenoma.^{206–208} Three independent sequencing studies suggested that L168R, T158A, and G151R mutations in *KCNJ5* (encodes inward-rectifier potassium channels

subunit 3.4) caused hypertension via enhanced aldosterone production.^{206–208} The authors also demonstrated that mutations at different regions of *KCNJ5*, such as G151E caused a milder phenotype, indicating site-specific importance in causing hypertensive phenotype.

Genome-Wide Association Studies

Previous family and twin studies have shown that genetic factors contribute to 30% to 50% of blood pressure variation.²⁰⁹ Genome-wide association studies focused on hypertension and blood pressure traits have identified >1000 genetic loci associated with increased susceptibility over the past 2 decades.^{210,211} These findings highlight the complex genetic landscape underlying blood pressure regulation.

One study investigated blood pressure traits, including systolic blood pressure, diastolic blood pressure, and pulse pressure, in 217 562 cases and 191 389 controls from the UK Biobank and the International Consortium of Blood Pressure Genome-Wide Association Studies. This study identified 3 distinct single-nucleotide polymorphisms (SNPs) within the introns of *KCNE2*, *KCNQ5*, and *KCNB1*, which encode K_v subfamily E member 2, K_v7.5, and K_v2.1 channels, respectively.²¹⁰ Five independent studies analyzed DNA sequence variants across 4 distinct biobank data sets, including the United Kingdom, Taiwan, European, and East Asia/Japanese, involving a total of 283 505 cases and 231 850 controls. These analyses led to the identification of multiple genetic loci associated with hypertension risk, such as a SNP within the intron of the *KCNK3*, which encodes TWIK-related acid-sensitive

potassium channel 1, a member of the 2-pore domain potassium channel (Table 6).^{212–216} No functional testing has been performed on any of the genome-wide association studies–identified SNPs, so their potential impact on potassium channel transcript levels, protein levels, or function is still unknown. Most of the SNPs listed in Table 6 are intronic variants, and one is intergenic, with none occurring in coding regions or causing amino acid changes. Therefore, if these SNPs have a functional effect, it would likely be due to their location in noncoding regulatory regions (such as enhancers, microRNAs, or long noncoding RNAs) that alter transcript levels, rather than directly affecting protein activity.

K⁺ CHANNELS AND ANTIARRHYTHMIC DRUGS IN THE HYPERTENSION CONTEXT

According to the extension of Vaughan Williams classification of antiarrhythmic drugs, class III antiarrhythmic drugs target potassium channels to help restore normal heart rhythm and exhibit blood pressure–lowering effects, highlighting the important link between potassium channel function and cardiovascular pharmacotherapy.^{9,14,217} Amiodarone, dronedarone, sotalolol, dofetilide, ibutilide, vernakalant, and nicorandil are the most frequently used class III antiarrhythmic drugs.^{9,218}

Amiodarone

Amiodarone is a potent antiarrhythmic drug used to treat AF, atrial flutter, and ventricular tachycardia.²¹⁹ Primarily, amiodarone prolongs the cardiac AP and refractory period

by blocking potassium channels (class III effect).²²⁰ It also inhibits inward sodium currents (class I effect),²²¹ which slows conduction velocity; weakly blocks beta-adrenergic receptors (class II effect), reducing sympathetic stimulation²²² and further slowing conduction through the atrio-ventricular node.^{223,224} By stabilizing abnormal electrical activity across these pathways, amiodarone reduces the likelihood of both atrial and ventricular arrhythmias.²¹⁹

A preclinical study in SHR has demonstrated that amiodarone treatment is associated with significantly reduced mean arterial pressure, showing values of 135±8 mmHg compared with 187±8 mmHg in control SHR (*P*<0.05), increased baroreflex sensitivity, diminished vasomotor sympathetic modulation, and decreased cardiac hypertrophy.²²⁵ Cushing et al²²⁶ demonstrated that intravenous amiodarone induced a sustained hypotensive effect throughout a period of 6 hours, marked by rapid declines in mean aortic pressure and cardiac output during continuous infusion in anesthetized beagle dogs. Interestingly, this hypotensive response was absent when a cosolvent-free formulation was used, suggesting that standard cosolvents contributed to the observed hypotension rather than amiodarone alone.²²⁶ These findings support the potential repurposing of amiodarone as an antihypertensive agent through its vasodilatory and negative inotropic effects, although formulation optimization is necessary to separate therapeutic blood pressure reduction from formulation-related side effects.²²⁶

In human studies, changes in blood pressure have mainly been reported as secondary outcomes in patients receiving amiodarone for arrhythmias or heart failure. A prospective study (Unique identifier: ChiCTR2200057816; URL: <https://www.chictr.org.cn/>

Table 6. SNPs in Potassium Ion Channel Genes Associated With Hypertension in Genome-Wide Association Studies

Strongest SNP-risk allele	SNPs	Mapped gene	Nearest gene	Ethnicity	Mapped trait	Population size		Male:female, %	OR (per unit/mm Hg) [95% CI]	P value	Con-text	Refer-ences
						Case	Control					
rs9976596-T	rs9976596	LINC00310-KCNE2	AP000320.7	European	Pulse pressure	217 562	191 389	45.26:54.74	0.19 [0.14–0.23]	2.00×10 ^{−14}	Intron variant	210
rs12195276-T	rs12195276	KCNQ5	KCNQ5	European	Pulse pressure			45.26:54.74	0.19 [0.15–0.22]	2.00×10 ^{−21}	Intron variant	
rs237485-A	rs237485	KCNB1	KCNB1	European	Diastolic blood pressure			45.25:54.75	0.11 [0.08–0.15]	4.00×10 ^{−9}	Intron variant	
rs2586886-?	rs2586886	KCNK3	NR	British	Hypertension	154 743	30 822	45.90:54.10	0.04 [NR]	4.00×10 ^{−9}	Intron variant	212
rs1275979-?	rs1275979	KCNK3	NR	Taiwanese	Hypertension	9653	59 333	38.08:61.92	0.17 [NR]	3×10 ^{−17}	Intron variant	213
rs1275923-T	rs1275923	KCNK3	KCNK3	European	Hypertension	28 391	28 246	42.00:58.00	0.90 [NR]	6×10 ^{−14}	Intron variant	214
rs1731243-T	rs1731243	KCNK3	KCNK3, AX746487	East Asian/Japanese	Hypertension	22 566	28 226	45.73:54.27	0.12 [0.09–0.15]	2×10 ^{−18}	Intron variant	215
rs1731249-A	rs1731249	KCNK3	NR	Japanese	Hypertension	22 566	28 226	45.73:54.27	0.12 [0.09–0.15]	5×10 ^{−14}	Intron variant	216
rs1275988-T	rs1275988	RPL37P11-KCNK3	NR	Japanese	Hypertension	22 566	28 226	45.73:54.27	0.18 [0.09–0.15]	1×10 ^{−13}	Intron variant	
rs35021474-C	rs35021474	KCNK3	NR	Japanese	Hypertension	22 566	28 226	45.73:54.27	0.13 [0.10–0.17]	2×10 ^{−13}	Intron variant	

NR indicates not reported; OR, odds ratio; and SNP, single-nucleotide polymorphism.

searchproj.html?regno=ChiCTR2200057816) involving 242 patients with heart failure with reduced ejection fraction demonstrated improved heart rate control (64.5 ± 3.2 bpm) and enhanced cardiac function with amiodarone treatment; blood pressure remained stable, with no clearly reported clinically significant difference in systolic blood pressure between groups in this study.²²⁷ In contrast, an observational study of chronic oral amiodarone use in 67 patients reported mild reductions in blood pressure compared with baseline readings as part of routinely documented clinical side effects, noting hemodynamic changes consistent with earlier findings on its vascular actions.²²⁸ Although not the main end point, the study noted clinically relevant hemodynamic changes consistent with previous data on amiodarone's effects on vascular tone.²²⁸ Despite these observations, amiodarone is not recommended for routine hypertension management. Its use is constrained by adverse effects such as thyroid dysfunction (notably fetal hypothyroidism during pregnancy), bradycardia, hepatic toxicity, and pulmonary effects,²²⁹ with higher risk in older patients or those with hepatic impairment.²³⁰ The observed hypotensive effects are thought to result from vasodilation, autonomic modulation, and adrenergic blockade. Consequently, any potential antihypertensive benefit is considered incidental, and use remains limited to cases where arrhythmia coexists.

Sotalol

Sotalol works as an antiarrhythmic by prolongation of the cardiac AP and refractory period.²³¹ Sotalol blocks the rapid delayed rectifier potassium channels in cardiac muscle cells (a class III property), which prolong repolarization and reduce the chance of ventricular tachycardia or AF.^{231,232} In addition, sotalol blocks beta-adrenergic receptors throughout the heart and blood vessels (a class II effect).²³¹ By dampening the influence of adrenaline and noradrenaline, it reduces heart rate,²³³ slows conduction through the atrioventricular node, and decreases the force of contraction.²³¹

Sotalol showed mild hypotensive effects in a study using cynomolgus monkeys, with an approximate 10% to 20% reduction in blood pressure observed at higher doses; however, this effect did not reach statistical significance across standard dosing intervals, likely due to the small group size ($n=4$).²³³ It is believed that sotalol's main cardiovascular effect is bradycardia due to beta blockade.²³³ An antihypertensive effect of sotalol has primarily been demonstrated in randomized controlled and comparative trials, where it is often studied alongside other beta-blockers. Leary et al conducted a randomized controlled trial in 25 patients with hypertension and found that single daily doses of sotalol (160–800 mg) effectively controlled elevated blood pressure for 24 hours, with minimal side effects.²³⁴ Andersson et al²³⁵ performed a double-blind, crossover randomized control trial comparing sotalol and metoprolol, showing both drugs produced clinically useful

reductions in blood pressure as monotherapies or combined with thiazide diuretics. However, sotalol requires dose adjustment or avoidance in patients with reduced renal function (creatinine clearance <40 mL/min),²³⁶ and adverse effects such as bradycardia, bronchospasm, and QT interval prolongation must be monitored.²³⁷

Although sotalol blocks cardiac potassium channels, the blood pressure-lowering effect appears to be driven mainly by β -blockade rather than potassium channel modulation. A similar limitation in mechanistic understanding applies to amiodarone. Although both drugs interact with multiple potassium channel subtypes,^{9,238,239} it remains unclear which specific channels contribute to their vascular effects. Targeted mechanistic studies, particularly using genetic knockout or selective pharmacological modulators, are needed to determine whether potassium channel modulation plays a meaningful role in their antihypertensive actions. In addition to the existing class III antiarrhythmic drugs, there are emerging therapies that attempt to selectively target SK_{Ca} (SK2 and SK3) and $K_v1.5$ (I_{Kur}) channels as discussed below.

Small-Conductance Calcium-Activated K^+ Channels (SK2 and SK3) Inhibitors

Pharmacological inhibition of SK_{Ca} channels has shown variable effects in AF models. Atrial refractoriness refers to the short period after an AP during which atrial tissue is temporarily unresponsive to re-excitation, thereby limiting premature impulses and reentry.⁸ In a rat AF model, pore-blocking agents such as ditrifluoroacetate hydrate (UCL1684) reduced AF duration compared with the vehicle-treated group ($23.38\% \pm 8.8\%$, $P<0.0001$) and increased atrial effective refractory period ($148.70 \pm 37.16\%$, $P<0.05$), a functional measure of atrial refractoriness.²⁴⁰ Icagen inhibited SK_{Ca} channels and prolonged APD in atrial myocytes from sinus rhythm patients.²⁴¹ Similarly, the Ca^{2+} -Calmodulin sensor modulators, NS8593 and AP14145 demonstrated antiarrhythmic actions. NS8593 reduced AF duration in rats and increased APD in human atrial myocytes,^{240,241} whereas AP14145 prolonged atrial refractoriness and reduced AF duration in both pig and guinea pig AF models.^{242,243} However, SK2 channel knockout in mice increased AF susceptibility despite APD prolongation,⁹ underscoring the dual antiarrhythmic and proarrhythmic potential of SK_{Ca} channel modulation. Together, these findings suggest that SK_{Ca} channel blockade via either pore blockers or Ca^{2+} -Calmodulin modulators can influence atrial electrophysiology and arrhythmia vulnerability in a context-dependent manner (Table 7).

Ultrarapid Delayed Rectifier K^+ Channels ($K_v1.5$) Blockers

$K_v1.5$ (I_{Kur}) channel blockers have been widely investigated as therapeutic options for AF. Vernakalant

Table 7. Evidence of K⁺ Channel Modulating Drugs on Improving Action Potential Characteristics in the Cardiac System

Target channel	Target current	Drug name (dose)	In vitro/in vivo evidence	Clinical trial evidence	Key findings	References
K _{Ca}	I _{KCa} – SK1–3 blockers	UCL1684	AF rat model	...	↓ AF duration (23.38% ±8.8%, <i>P</i> <0.0001), ↑ aERP (148.70%±37.16%, <i>P</i> <0.05)	240
		Icagen	Human atrial myocytes and trabeculae	...	↓ I _{Kr} ↑ APD and ERP (SR patients only)	241
		NS8593	Rat model and human atrial myocytes	...	↓ AF duration (64.5%), ↑ APD in SR myocytes	240,241
		AP14145	Vernakalant-resistant pig AF and atrial paced in vivo guinea pig model	...	↑ Atrial refractoriness, ↓ AF duration, no ventricular effects	242,243
K _v	I _{Kur} blockers	Vernakalant	Transfected HEK293 cells, canine and human atrial myocytes	Phases III/IV (URL: https://www.clinicaltrials.gov ; Unique identifiers: NCT04485195; NCT00989001; NCT03005366; NCT04748991)	↑ AF conversion rate (51.5% vs 3.8% placebo), median time to conversion 11 min	9,143,244–247
		AVE0118	Human atrial myocytes and trabeculae	...	↓ Outward K ⁺ current, differences between SR and AF tissue	248
		Acacetin	Human atrial myocytes	...	Suppressed I _{Kur} current and prolonged APD	249
		MK-0448	Human atrial tissues	...	↑ ERP in atrial tissue	250
		XEN D0101	Human embryonic stem cell-derived atrial cardiomyocytes, human atrial myocytes, and atrial trabeculae	Phase 1 (healthy male volunteers)	Safe, dose-dependent effects on atrial electrophysiology	9,251,252
		XEN D0103	Human atrial myocytes (SR, permanent AF, and paroxysmal AF)	Phase I (URL: https://www.clinicaltrials.gov ; Unique identifier: NCT02390258)	Safe, dose-dependent corrected QT interval.	9,253
		DPO-1	Transfected CHO cells, human atrial myocytes, and guinea pig ventricular myocytes	...	↓ I _{Kur} , prolonged APD	254
		Ambasilide	Canine vagotonic AF model	...	↓ AF duration	255
		DHA and EPA	Adult human atrial myocytes	Phases III/IV (URL: https://www.clinicaltrials.gov ; Unique identifiers: NCT00791089; NCT00970489; NCT01175330; NCT01198275; NCT00402363; NCT03078738)	↓ I _{Kur}	9,256
		Amiodarone	In vivo dog, horse, pig, and goat models	Phase IV (URL: https://www.clinicaltrials.gov ; Unique identifiers: NCT04594746; NCT03432663; NCT00845780; NCT02341105)	Prolonged aERP and aERP adaptation	9,238
		BMS-919373	Rabbit aERP model	Phase II (URL: https://www.clinicaltrials.gov ; Unique identifiers: NCT02156076)	↓ I _{Kur} and prolonged aERP (9.0±2.4%)	9,257

A-935142 indicates 4-[4-(5-trifluoromethyl-1H-pyrazol-3-yl)-phenyl]-cyclohexyl-acetic acid; aERP, atrial effective refractory period; AF, atrial fibrillation; APD, action potential duration; AZSMO-23, N-[3-(1H-benzimidazol-2-yl)-4-chloro-phenyl]pyridine-3-carboxamide; BMS-919373, (Phenyl-*N*-(pyridin-2-ylmethyl)-2-(pyrimidin-5-yl)quinazolin-4-amine); CHO, Chinese hamster ovary; DHA, docosahexaenoic acid; DPO-1, diphenyl phosphide oxide; EPA, eicosapentanoic acid; ERP, effective refractory period; HEK293 cells, human embryonic kidney cells; hERG, Human ether-a-go-go-related gene (hERG) potassium channel; ICA-105574, (3-Nitro-*N*-(4-phenoxyphenyl)Benzamide); I_{KCa}, intermediate-conductance calcium-activated potassium channel; I_{Kur}, ultrarapid delayed rectifier potassium current; K⁺, potassium ion; K_v, voltage-gated potassium channel; K_v1.5, voltage-gated potassium channel encoded by the *KCNK5* gene; MK-0448, (N-6-[(1*S*)-1-(4-fluorophenyl)-2,2-di(pyridin-3-yl)ethyl]pyridin-2-ylmethanesulfonamide); ML277, (R)-*N*-(4-(4-methoxyphenyl)thiazol-2-yl)-1-tosylpiperidine-2-carboxamide; NS1643, 1,3bis-(2-hydroxy-5-trifluoromethylphenyl)-urea; PD-118057, (2-(4-[2-(3,4-dichloro-phenyl)-ethyl]phenylamino)-benzoic acid); RPR260243, (3*R*,4*R*)-4-[3-(6-methoxyquinolin-4-yl)-3-oxo-propyl]-1-[3-(2,3,5-trifluoro-phenyl)-prop-2-ynyl]-piperidine-3-carboxylic acid; and SR, sinus rhythm.

demonstrated conversion of acute AF to sinus rhythm in both preclinical and Phase III/IV clinical studies,^{143,244–247} although its electrophysiological effects varied between species. However, it was found to be nonselective, affecting not only I_{Kur} but also voltage-gated potassium channels encoded by the *KCN2D* gene, voltage-gated

potassium channels encoded by the *KCNH2* gene, inward-rectifier potassium channels subunit 3.1/inward-rectifier potassium channels subunit 3.4, K_{ir}6.1, K_{ir}6.2, and late sodium currents.^{244,258,259} Several other agents, including AVE0118,²⁴⁸ Acacetin,²⁴⁹ and MK-0448,²⁵⁰ prolonged APD or refractory period in human atrial myocytes, while

XEN D0101^{251,252} and XEN D0103²⁵³ showed early phase safety and pharmacological effects in clinical trials. Additionally, compounds such as diphenyl phosphide oxide-1,²⁵⁴ ambasilide,²⁵⁵ and omega-3 fatty acids (docosahexaenoic acid/eicosapentanoic acid)²⁵⁶ have also exhibited modulatory effects on atrial currents, with some progressing to late-stage clinical evaluation. Classic antiarrhythmic agents such as amiodarone remain effective in prolonging atrial ERP and suppressing AF, but are nonselective.²³⁸ Furthermore, BMS-919373, a potent and selective I_{Kur} inhibitor, has shown promising preclinical efficacy and entered Phase II trials.²⁵⁷ Collectively, these findings indicate that I_{Kur} inhibition can prolong atrial refractoriness and suppress AF, although clinical translation has produced mixed outcomes (Table 7).

Together, these findings highlight the potential inhibition of atrial-specific repolarizing currents (I_{Kur} , SK) to prolong refractoriness and suppress AF. Furthermore, both animal and human studies suggest that downregulation of SK and I_{Kur} currents may have potential blood pressure-lowering effects. However, blockade of endothelial SK3 channels depolarizes ECs and attenuates NO/EDH-mediated vasodilation,²⁶⁰ while inhibition of $K_v1.5$ channels in VSMCs reduces metabolism-driven dilation and shifts vascular tone toward constriction and impaired blood flow.⁶⁵ The net blood pressure effect will therefore depend on the balance between central cardiac effect and vascular effects, highlighting the importance of cardiac-selective targeting to avoid unintended hypertension.

FUTURE DIRECTIONS

Future research should aim to bridge the knowledge gaps in K^+ channel function and its role in cardiovascular diseases by integrating findings from human and rodent studies. These comparative studies are crucial for identifying species-specific differences and developing humanized models to test hypotheses. The changes observed in BK_{Ca} , K_v , and K_{ir} channel activity in hypertensive rodent models should be validated in diverse human populations with varying ethnicities, age, and multiple comorbidities.

Functional assays are needed to characterize the impact of genome-wide association studies-identified SNPs, particularly those SNPs in regulatory regions of K^+ channel expression, which could significantly affect vascular function. Similarly, investigating somatic mutations in KCNJ5, such as L168R, T158A, and G151R, can enhance understanding of their role in the aldosterone-driven cause of hypertension using improved clinically representative animal models or human tissue samples offering valuable insights. Research focusing on early changes in K^+ channel function in young hypertensive individuals and tracking its progression will provide insights into the role of K^+ channels during disease development. Additionally, exploring the role of K^+ channels in secondary hypertension, particularly in conditions

like primary aldosteronism or renal artery stenosis, can uncover therapeutic opportunities.

An important challenge for translating K^+ channel biology into antihypertensive therapy is the marked heterogeneity of channel expression and function across vascular beds. For example, mesenteric,^{80,81,88,176,181,183,184} cerebral,^{175,177,179,183,185} thoracic,^{82,182,183} carotid¹⁷⁸ and renal¹⁸³ arteries differ in their reliance on BK_{Ca} ,^{80–82,175–179,181,182} K_v ,^{82,88,183} K_{ATP} ,¹⁸⁴ and K_{ir} ,^{184,185} channels for setting resting tone, flow-mediated responses and protection of tissue perfusion under stress. This implies that global K^+ channel-directed approach may be suboptimal, particularly when specific target organs (eg, brain versus heart versus kidney) are at greatest risk. Future work should therefore aim to define vascular bed-specific channel fingerprints in hypertension and major comorbidities. Such information could guide therapeutic strategies.

Pharmacological agents targeting specific potassium channel subunits, such as $BK_{Ca}\beta1$ or $K_v7.4$, hold potential for the treatment of cardiovascular diseases affecting blood vessel tone. Ideally, next-generation drugs or delivery systems would exploit potassium channel distribution heterogeneity (eg, via tissue-selective compounds or targeted delivery) to maximize blood pressure reduction and organ protection while minimizing off-target effects in vascular beds where a given channel plays a different role. Studies summarized in Table 4 were performed in established hypertensive models, meaning that mechanistic data for clearly defined early, intermediate, and late stages are scarce. Future work should therefore use longitudinal or age-stratified designs within the same animal model to map how specific vascular K^+ channel subtypes (such as BK_{Ca} , K_v , K_{ATP} , K_{ir}) change from prehypertensive through advanced disease. Such stage-resolved description of channel expression and function would provide a rational basis for developing targeted K^+ channel-directed interventions tailored to early versus late hypertension, rather than assuming a uniform channel profile across all stages.

Notably, several existing antiarrhythmic drugs that act on potassium currents (such as I_{Kur} blockers and SK channel blockers) provide a strong basis for potential strategic drug repurposing. Future research should explore whether these agents, originally developed for arrhythmia treatment, could also be used to regulate vascular tone and improve outcomes in conditions like peripheral artery disease.

In addition, transcriptomic and epigenomic studies using techniques like single-cell RNA sequencing are needed to investigate regulatory mechanisms affecting potassium channel gene expression, offering a cell type-specific perspective. Longitudinal clinical studies are also important to understand the progression of potassium channel dysfunction in patients with hypertension and evaluate the impact of interventions like exercise and diet.

Most work to date has focused on small-molecule openers or blockers of vascular K^+ channels, but newer

modalities may eventually allow more precise manipulation of channel expression and function in hypertension. Experimental studies in hypertension rat models show that adeno-associated virus-mediated delivery of vasodilator genes, such as atrial natriuretic peptide, kallikrein, adrenomedullin, and eNOS, results in a lowering of blood pressure.^{261–266} In the same way, delivery of K⁺ channel genes through adeno-associated virus targeting blood vessels can achieve relatively stable, tissue-selective overexpression, raising the possibility of restoring K⁺ channel subunits (eg, BK_{Ca}β1 or K_v7 family members) in resistance arteries to counteract pathological depolarization and increased tone. Conversely, RNA-based approaches such as small interfering RNA²⁶⁷ or antisense oligonucleotides²⁶⁸ could be used to selectively knock down overactive or maladaptive channel subtypes in specific vascular beds, potentially with endothelial or smooth muscle cell targeted delivery systems.^{269,270} Collectively, these research directions would deepen our understanding of the complex role of potassium channels in cardiovascular conditions such as hypertension and aid in the development of targeted therapeutic strategies.

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Author Contribution

T. Madiwila Gamarachchige performed data extraction and prepared the initial draft. S. Thanigaimani and K.S. Barratt were both involved in data extraction and reviewing the draft. J. Golledge was involved in conceptualization, writing and editing the review, supervision, and funding. All authors were involved in providing critical revisions to the manuscript.

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